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United States Patent Application Publication

20250276765

Kind Code

A1

Publication Date

September 04, 2025

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WASTE CONTAINER AND VALVE APPARATUS THEREFOR

Abstract

Aspects of the present disclosure provide waste containers for waste management on a vehicle, such as a watercraft. A waste container for use on a vehicle may include at least one wall comprising an inner surface defining a chamber for holding waste, and a plurality of projections projecting from the inner surface of the at least one wall into the chamber. Aspects of the present disclosure also provide valve apparatuses for controlling flow of waste into or out of a waste container. Such a valve apparatus may include a non-dismantlable valve body defining a flow channel for conveying waste, and at least one check valve secured within the flow channel and configured to allow waste to flow from an inlet to an outlet of the flow channel and to prevent waste from flowing from the outlet to the inlet.

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Family ID: 96880882

Appl. No.: 19/066813

Filed: February 28, 2025

Related U.S. Application Data

us-provisional-application US 63561084 20240304

us-provisional-application US 63656142 20240605

Publication Classification

Int. Cl.: B63B17/06 (20060101); B63B29/14 (20060101); B63B29/16 (20060101); B63B73/20 (20200101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims the benefit of, and priority to, U.S. provisional patent application No. 63/561,084 filed Mar. 4, 2024, the entire contents of which are incorporated by reference herein. This application also claims the benefit of, and priority to, U.S. provisional patent application No. 63/656,142 filed Jun. 5, 2024, the entire contents of which are incorporated by reference herein.

FIELD

[0002] This disclosure relates generally to containers and valves for waste management on a vehicle, such as a watercraft.

BACKGROUND

[0003] Vehicles such as watercraft may include waste containers, valves, and pumps for storing and managing waste onboard. However, over time, solid wastes may accumulate in a waste container and may block routes for emptying the container. Additionally, maintenance of valves in existing onboard waste management systems may be overly complex.

SUMMARY

[0004] According to at least one embodiment, there is disclosed a waste container for use on a vehicle, the waste container comprising: at least one wall comprising an inner surface defining a chamber for holding waste; and a plurality of projections projecting from the inner surface of the at least one wall into the chamber.

[0005] In some embodiments, the projections are configured to disturb flow of waste in the chamber as movement of the vehicle causes the waste in the chamber to move in the chamber.

[0006] In some embodiments, the projections are configured to generate turbulence in the waste in the chamber as the movement of the vehicle causes the waste in the chamber to move in the chamber.

[0007] In some embodiments, the projections are configured to break apart solids in the waste in the chamber as the movement of the vehicle causes the waste in the chamber to move in the chamber.

[0008] In some embodiments, the projections are configured to disperse particles in the waste in the chamber as the movement of the vehicle causes the waste in the chamber to move in the chamber.

[0009] In some embodiments, the movement of the vehicle comprises tilting of the vehicle.

[0010] In some embodiments, at least one of the plurality of projections extends along the inner surface without contacting any other one of the plurality of projections.

[0011] In some embodiments, each of the plurality of projections extends along the inner surface without contacting any other one of the plurality of projections.

[0012] In some embodiments, at least some of the plurality of projections extend along the inner surface parallel to one another.

[0013] In some embodiments, adjacent ones of the plurality of projections along the inner surface, which extend parallel to one another along the inner surface, are spaced apart from one another at a fixed projection spacing in a spacing direction along the inner surface, the spacing direction perpendicular to an extension direction along the inner surface of the adjacent ones of the plurality of projections.

[0014] In some embodiments, the projection spacing is at most 1 inch.

[0015] In some embodiments, the projection spacing is about 1 inch.

[0016] In some embodiments, the projection spacing is at least 1 inch.

[0017] In some embodiments, the projection spacing is at most 2 inches.

[0018] In some embodiments, the projection spacing is about 2 inches. In some embodiments, each of the plurality of projections extends along the inner surface in a substantially vertical or vertical direction when the waste container is installed on the vehicle.

[0019] In some embodiments, each of the plurality of projections projects from a respective portion of the inner surface in a respective projection direction generally orthogonal or orthogonal to the respective portion of the inner surface.

[0020] In some embodiments, at least a portion of each of the plurality of projections projects at least 0.5 inches from the inner surface.

[0021] In some embodiments, all of each of the plurality of projections projects at least 0.5 inches from the inner surface.

[0022] In some embodiments, at least a portion of each of the plurality of projections projects about 0.5 inches from the inner surface.

[0023] In some embodiments, all of each of the plurality of projections projects about 0.5 inches from the inner surface.

[0024] In some embodiments, at least a portion of each of the plurality of projections projects at least 1.25 inches from the inner surface.

[0025] In some embodiments, all of each of the plurality of projections projects at least 1.25 inches from the inner surface.

[0026] In some embodiments, at least a portion of each of the plurality of projections projects about 1.25 inches from the inner surface.

[0027] In some embodiments, all of each of the plurality of projections projects about 1.25 inches from the inner surface.

[0028] In some embodiments, each of the plurality of projections projects at most 0.5 inches from the inner surface.

[0029] In some embodiments, each of the plurality of projections projects at most 1.25 inches from the inner surface.

[0030] In some embodiments, the at least one wall further defines at least one opening in the chamber for conveying waste into and out of the chamber.

[0031] In some embodiments, the at least one opening is closable.

[0032] In some embodiments, at least one of the at least one opening is at a top of the chamber or a bottom of the chamber when the waste container is installed on the vehicle.

[0033] In some embodiments, the at least one opening comprises: at least one inlet for receiving waste into the chamber; and at least one outlet for removing waste from the chamber.

[0034] In some embodiments, the at least one outlet comprises a bottom outlet positioned to be at the bottom of the chamber when the waste container is installed on the vehicle.

[0035] In some embodiments, the waste container further comprises a valve operable to reversibly close the bottom outlet.

[0036] In some embodiments, the valve is a ball valve.

[0037] In some embodiments, the bottom outlet is configured to be in fluid communication with a seacock on a hull of the vehicle when the waste container is installed on the vehicle.

[0038] In some embodiments, the at least one outlet comprises a top outlet positioned to be at the top of the chamber when the waste container is installed on the vehicle.

[0039] In some embodiments, the waste container further comprises a dip tube extending from the top outlet into the chamber, the dip tube in fluid communication with the top outlet and with the chamber.

[0040] In some embodiments, the top outlet is configured to be in fluid communication with a pump-out fitting of the vehicle when the waste container is installed on the vehicle.

[0041] In some embodiments, the at least one inlet comprises a top inlet positioned to be at the top of the chamber when the waste container is installed on the vehicle.

[0042] In some embodiments, the at least one inlet is configured to be in fluid communication with a toilet onboard the vehicle when the waste container is installed on the vehicle.

[0043] In some embodiments, the at least one inlet is configured to be in fluid communication with a sink onboard the vehicle when the waste container is installed on the vehicle.

[0044] In some embodiments, the at least one inlet is configured to be in fluid communication with a shower onboard the vehicle when the waste container is installed on the vehicle.

[0045] In some embodiments, each of the plurality of projections extends along at least a portion of the inner surface between one of the at least one inlet and one of the at least one outlet.

[0046] In some embodiments, the at least one wall comprises a plurality of walls.

[0047] In some embodiments, at least some of the plurality of walls are parallel to one another.

[0048] In some embodiments, at least one of the plurality of projections projects from each of the plurality of walls.

[0049] In some embodiments, none of the plurality of projections project from at least one of the plurality of walls.

[0050] In some embodiments, the vehicle is a marine vessel.

[0051] In some embodiments, the waste container is installed on the vehicle.

[0052] According to at least another embodiment, there is disclosed a method of manufacturing the waste container, the method comprising molding the at least one wall with the plurality of projections.

[0053] According to at least another embodiment, there is disclosed a method of manufacturing the waste container, the method comprising: molding the at least one wall; and fixing the plurality of projections to the inner surface.

[0054] In some embodiments, molding the at least one wall comprises rotational molding the at least one wall.

[0055] In some embodiments, molding the at least one wall comprises injection molding the at least one wall.

[0056] According to at least another embodiment, there is disclosed a system comprising: the waste container; and the vehicle, wherein the waste container is installed on the vehicle.

[0057] According to at least another embodiment, there is disclosed a use of the waste container for waste management on the vehicle.

[0058] According to at least another embodiment, there is disclosed a valve apparatus for controlling flow of waste into or out of a waste container, the valve apparatus comprising a non-dismantlable valve body defining a flow channel for conveying the waste, the flow channel comprising a channel inlet for receiving the waste into the flow channel and a channel outlet for removing the waste from the flow channel; and at least one check valve secured within the flow channel, the at least one check valve configured to: allow the waste to flow through the flow channel in a forward flow direction from the channel inlet to the channel outlet; and prevent the waste from flowing through the flow channel in a reverse flow direction from the channel outlet to the channel inlet.

[0059] In some embodiments, the valve apparatus is for use with a pump for controlling the flow of the waste into or out of the waste container.

[0060] In some embodiments, the pump comprises a vacuum pump.

[0061] In some embodiments, the valve body comprises an inlet end defining the channel inlet, the inlet end connectable to a waste source to cause the flow channel to be in fluid communication with the waste source through the channel inlet.

[0062] In some embodiments, the valve apparatus further comprises a channel inlet clamp operable to releasably secure the inlet end to the waste source to connect the inlet end to the waste source to cause the flow channel to be in fluid communication with the waste source through the channel

inlet.

[0063] In some embodiments, the inlet end comprises at least one inlet end interlocking portion, the at least one inlet end interlocking portion releasably interlockable with a corresponding waste source interlocking portion of the waste source to releasably secure the inlet end to the waste source to connect the inlet end to the waste source to cause the flow channel to be in fluid communication with the waste source through the channel inlet.

[0064] In some embodiments, the at least one inlet end interlocking portion is releasably interlockable with the waste source interlocking portion by rotation of the valve apparatus relative to the waste source.

[0065] In some embodiments, the rotation of the valve apparatus relative to the waste source is around an inlet channel axis parallel to the flow channel at the channel inlet.

[0066] In some embodiments, the at least one inlet end interlocking portion projects from the valve body at the inlet end.

[0067] In some embodiments, the at least one inlet end interlocking portion projects out from the valve body away from the flow channel.

[0068] In some embodiments, the inlet end comprises an inlet end mating surface around the channel inlet, the inlet end mating surface configured to engage with a corresponding waste source mating surface of the waste source when the inlet end is connected to the waste source to cause the flow channel to be in fluid communication with the waste source through the channel inlet.

[0069] In some embodiments, the valve apparatus further comprises a channel inlet seal seated on the inlet end mating surface around the channel inlet, the channel inlet seal configured to seal an interface between the inlet end mating surface and the waste source mating surface when the inlet end mating surface engages with the waste source mating surface.

[0070] In some embodiments, the channel inlet seal comprises a face seal.

[0071] In some embodiments, the channel inlet seal comprises an O-ring.

[0072] In some embodiments, the waste source comprises the waste container.

[0073] In some embodiments, the waste source comprises the vacuum pump.

[0074] In some embodiments, the valve body comprises an outlet end defining the channel outlet, the outlet end connectable to a waste destination to cause the flow channel to be in fluid communication with the waste destination through the channel outlet.

[0075] In some embodiments, the valve apparatus further comprises a channel outlet clamp operable to releasably secure the outlet end to the waste destination to connect the outlet end to the waste destination to cause the flow channel to be in fluid communication with the waste destination through the channel outlet.

[0076] In some embodiments, the outlet end comprises at least one outlet end interlocking portion, the at least one outlet end interlocking portion releasably interlockable with a corresponding waste destination interlocking portion of the waste destination to releasably secure the outlet end to the waste destination to connect the outlet end to the waste destination to cause the flow channel to be in fluid communication with the waste destination through the channel outlet.

[0077] In some embodiments, the at least one outlet end interlocking portion is releasably interlockable with the waste destination interlocking portion by rotation of the valve apparatus relative to the waste destination.

[0078] In some embodiments, the rotation of the valve apparatus relative to the waste destination is around an outlet channel axis parallel to the flow channel at the channel outlet.

[0079] In some embodiments, the at least one outlet end interlocking portion projects from the valve body at the outlet end.

[0080] In some embodiments, the at least one outlet end interlocking portion projects out from the valve body away from the flow channel.

[0081] In some embodiments, the outlet end comprises an outlet end mating surface around the channel outlet, the outlet end mating surface configured to engage with a corresponding waste

destination mating surface of the waste destination when the outlet end is connected to the waste destination to cause the flow channel to be in fluid communication with the waste destination through the channel outlet.

[0082] In some embodiments, the valve apparatus further comprises a channel outlet seal seated on the outlet end mating surface around the channel outlet, the channel outlet seal configured to seal an interface between the outlet end mating surface and the waste destination mating surface when the outlet end mating surface engages with the waste destination mating surface.

[0083] In some embodiments, the channel outlet seal comprises a face seal.

[0084] In some embodiments, the channel outlet seal comprises an O-ring.

[0085] In some embodiments, the waste destination comprises the waste container.

[0086] In some embodiments, the waste destination comprises the vacuum pump.

[0087] In some embodiments, the valve body comprises a plurality of valve body portions between the channel inlet and the channel outlet, the plurality of valve body portions collectively defining the flow channel, each valve body portion of the plurality of valve body portions fixed to at least one other valve body portion of the plurality of valve body portions.

[0088] In some embodiments, at least two of the valve body portions of the plurality of valve body portions are fixed together by plastic welding.

[0089] In some embodiments, at least two of the valve body portions of the plurality of valve body portions are fixed together by glue.

[0090] In some embodiments, at least one valve body portion of the plurality of valve body portions retains the check valve.

[0091] In some embodiments, the at least one check valve comprises at least one duckbill valve.

[0092] In some embodiments, the at least one duckbill valve comprises a first duckbill valve and a second duckbill valve.

[0093] In some embodiments, the first duckbill valve and the second duckbill valve are non-overlapping in the forward flow direction.

[0094] In some embodiments, at least a portion of the first duckbill valve is received in the second duckbill valve.

[0095] In some embodiments, the first duckbill valve is nested within the second duckbill valve.

[0096] In some embodiments, the valve apparatus is for controlling the flow of the waste into or out of the waste container disclosed above.

[0097] According to at least another embodiment, there is disclosed a method of installing the valve apparatus to the waste source, the method comprising: causing the channel inlet clamp to secure the inlet end to the waste source.

[0098] According to at least another embodiment, there is disclosed a method of installing the valve apparatus to the waste source, the method comprising: rotating the valve apparatus relative to the waste source to interlock the at least one inlet end interlocking portion with the waste source interlocking portion.

[0099] According to at least another embodiment, there is disclosed a method of installing the valve apparatus to the waste destination, the method comprising: causing the channel outlet clamp to secure the outlet end to the waste destination.

[0100] According to at least another embodiment, there is disclosed a method of installing the valve apparatus to the waste destination, the method comprising: rotating the valve apparatus relative to the waste destination to interlock the at least one outlet end interlocking portion with the waste destination interlocking portion.

[0101] According to at least another embodiment, there is disclosed a system comprising: the valve apparatus; and the waste container, wherein: the waste container comprises at least one wall comprising an inner surface defining a chamber for holding the waste; and the flow channel of the valve apparatus is in fluid communication with the chamber of the waste container.

[0102] In some embodiments, the at least one wall of the waste container defines a container outlet

for removing the waste from the waste container; and the flow channel of the valve apparatus is in fluid communication with the chamber of the waste container through the container outlet of the waste container and through the channel inlet of the valve apparatus.

[0103] In some embodiments, the inlet end of the valve apparatus is connected to the waste container at the container outlet of the waste container.

[0104] In some embodiments, the channel inlet clamp of the valve apparatus releasably secures the inlet end of the valve apparatus to the waste container at the container outlet of the waste container.

[0105] In some embodiments, the waste container further comprises at least one waste container outlet interlocking portion at the container outlet, the at least one waste container outlet interlocking portion releasably interlocked with a corresponding one of the at least one inlet end interlocking portion of the inlet end of the valve apparatus to releasably secure the inlet end of the valve apparatus to the waste container at the container outlet of the waste container.

[0106] In some embodiments, the inlet end mating surface of the valve apparatus engages with a corresponding container outlet mating surface of the waste container.

[0107] In some embodiments, the channel inlet seal of the valve apparatus seals an interface between the inlet end mating surface of the valve apparatus and the container outlet mating surface of the waste container.

[0108] In some embodiments, the at least one wall of the waste container defines a container inlet for receiving the waste into the waste container; and the flow channel of the valve apparatus is in fluid communication with the waste container through the container inlet of the waste container and through the channel outlet of the valve apparatus.

[0109] In some embodiments, the outlet end of the valve apparatus is connected to the waste container at the container inlet of the waste container.

[0110] In some embodiments, the channel outlet clamp of the valve apparatus releasably secures the outlet end of the valve apparatus to the waste container at the container inlet of the waste container.

[0111] In some embodiments, the waste container further comprises at least one waste container inlet interlocking portion at the container inlet, the at least one waste container inlet interlocking portion releasably interlocked with a corresponding one of the at least one outlet end interlocking portion of the outlet end of the valve apparatus to releasably secure the outlet end of the valve apparatus to the waste container at the container inlet of the waste container.

[0112] In some embodiments, the outlet end mating surface of the valve apparatus engages with a corresponding container inlet mating surface of the waste container.

[0113] In some embodiments, the channel outlet seal of the valve apparatus seals an interface between the outlet end mating surface of the valve apparatus and the container inlet mating surface of the waste container.

[0114] In some embodiments, the waste container comprises the waste container disclosed above.

[0115] According to at least another embodiment, there is disclosed a vacuum pump system for controlling flow of waste into or out of a waste container, the vacuum pump system comprising: a vacuum pump operable to cause the waste to flow, the vacuum pump defining a pump inlet for receiving the waste into the vacuum pump and a pump outlet for removing the waste from the vacuum pump; and a first one of the valve apparatus disclosed above, wherein the flow channel of the first one of the valve apparatus is in fluid communication with the vacuum pump.

[0116] In some embodiments, the flow channel of the first one of the valve apparatus is in fluid communication with the vacuum pump through the pump outlet of the vacuum pump and through the channel inlet of the first one of the valve apparatus.

[0117] In some embodiments, the inlet end of the first one of the valve apparatus is connected to the vacuum pump at the pump outlet of the vacuum pump.

[0118] In some embodiments, the channel inlet clamp of the first one of the valve apparatus releasably secures the inlet end of the first one of the valve apparatus to the vacuum pump at the

pump outlet of the vacuum pump.

[0119] In some embodiments, the vacuum pump further comprises at least one vacuum pump outlet interlocking portion at the pump outlet, the at least one vacuum pump outlet interlocking portion releasably interlocked with a corresponding one of the at least one inlet end interlocking portion of the inlet end of the first one of the valve apparatus to releasably secure the inlet end of the first one of the valve apparatus to the vacuum pump at the pump outlet of the vacuum pump.

[0120] In some embodiments, the inlet end mating surface of the first one of the valve apparatus engages with a corresponding pump outlet mating surface of the vacuum pump.

[0121] In some embodiments, the channel inlet seal of the first one of the valve apparatus seals an interface between the inlet end mating surface of the first one of the valve apparatus and the pump outlet mating surface of the vacuum pump.

[0122] In some embodiments, the flow channel of the first one of the valve apparatus is in fluid communication with the vacuum pump through the pump inlet of the vacuum pump and through the channel outlet of the first one of the valve apparatus.

[0123] In some embodiments, the outlet end of the first one of the valve apparatus is connected to the vacuum pump at the pump inlet of the vacuum pump.

[0124] In some embodiments, the channel outlet clamp of the first one of the valve apparatus releasably secures the outlet end of the first one of the valve apparatus to the vacuum pump at the pump inlet of the vacuum pump.

[0125] In some embodiments, the vacuum pump further comprises at least one vacuum pump inlet interlocking portion at the pump inlet, the at least one vacuum pump inlet interlocking portion releasably interlocked with a corresponding one of the at least one outlet end interlocking portion of the outlet end of the first one of the valve apparatus to releasably secure the outlet end of the first one of the valve apparatus to the vacuum pump at the pump inlet of the vacuum pump.

[0126] In some embodiments, the outlet end mating surface of the first one of the valve apparatus engages with a corresponding pump inlet mating surface of the vacuum pump.

[0127] In some embodiments, the channel outlet seal of the first one of the valve apparatus seals an interface between the outlet end mating surface of the first one of the valve apparatus and the pump inlet mating surface of the vacuum pump.

[0128] In some embodiments, the vacuum pump system further comprises a second one of the valve apparatus disclosed above, wherein the flow channel of the second one of the valve apparatus is in fluid communication with the vacuum pump through the pump outlet of the vacuum pump and through the channel inlet of the second one of the valve apparatus.

[0129] In some embodiments, the inlet end of the second one of the valve apparatus is connected to the vacuum pump at the pump outlet of the vacuum pump.

[0130] In some embodiments, the channel inlet clamp of the second one of the valve apparatus releasably secures the inlet end of the second one of the valve apparatus to the vacuum pump at the pump outlet of the vacuum pump.

[0131] In some embodiments, the vacuum pump further comprises at least one vacuum pump outlet interlocking portion at the pump outlet, the at least one vacuum pump outlet interlocking portion releasably interlocked with a corresponding one of the at least one inlet end interlocking portion of the inlet end of the second one of the valve apparatus to releasably secure the inlet end of the second one of the valve apparatus to the vacuum pump at the pump outlet of the vacuum pump.

[0132] In some embodiments, the inlet end mating surface of the second one of the valve apparatus engages with a corresponding pump outlet mating surface of the vacuum pump.

[0133] In some embodiments, the channel inlet seal of the second one of the valve apparatus seals an interface between the inlet end mating surface of the second one of the valve apparatus and the pump outlet mating surface of the vacuum pump.

[0134] In some embodiments, the vacuum pump system is for controlling the flow of the waste into or out of the waste container disclosed above

[0135] According to at least another embodiment, there is disclosed a system comprising: the vacuum pump system as disclosed above; and the waste container, wherein: the waste container comprises at least one wall comprising an inner surface defining a chamber for holding the waste; and the vacuum pump system is in fluid communication with the chamber of the waste container.

[0136] In some embodiments, the flow channel of the first one of the valve apparatus of the vacuum pump system is in fluid communication with the chamber of the waste container.

[0137] In some embodiments, the at least one wall of the waste container defines a container inlet for receiving the waste into the waste container; and the flow channel of the first one of the valve apparatus of the vacuum pump system is in fluid communication with the chamber of the waste container through the container inlet of the waste container and through the channel outlet of the first one of the valve apparatus of the vacuum pump system.

[0138] In some embodiments, the outlet end of the first one of the valve apparatus of the vacuum pump system is connected to the waste container at the container inlet of the waste container.

[0139] In some embodiments, the channel outlet clamp of the first one of the valve apparatus of the vacuum pump system releasably secures the outlet end of the first one of the valve apparatus of the vacuum pump system to the waste container at the container inlet of the waste container.

[0140] In some embodiments, the waste container further comprises at least one waste container inlet interlocking portion at the container inlet, the at least one waste container inlet interlocking portion releasably interlocked with a corresponding one of the at least one outlet end interlocking portion of the outlet end of the first one of the valve apparatus to releasably secure the outlet end of the first one of the valve apparatus to the waste container at the container inlet of the waste container.

[0141] In some embodiments, the outlet end mating surface of the first one of the valve apparatus of the vacuum pump system engages with a corresponding container inlet mating surface of the waste container.

[0142] In some embodiments, the channel outlet seal of the first one of the valve apparatus of the vacuum pump system seals an interface between the outlet end mating surface of the first one of the valve apparatus of the vacuum pump system and the container inlet mating surface of the waste container.

[0143] In some embodiments, the at least one wall of the waste container defines a container outlet for receiving the waste into the waste container; and the flow channel of the first one of the valve apparatus of the vacuum pump system is in fluid communication with the chamber of the waste container through the container outlet of the waste container and through the channel inlet of the first one of the valve apparatus of the vacuum pump system.

[0144] In some embodiments, the inlet end of the first one of the valve apparatus of the vacuum pump system is connected to the waste container at the container outlet of the waste container.

[0145] In some embodiments, the channel inlet clamp of the first one of the valve apparatus of the vacuum pump system releasably secures the inlet end of the first one of the valve apparatus of the vacuum pump system to the waste container at the container outlet of the waste container.

[0146] In some embodiments, the waste container further comprises at least one waste container outlet interlocking portion at the container outlet, the at least one waste container outlet interlocking portion releasably interlocked with a corresponding one of the at least one inlet end interlocking portion of the inlet end of the first one of the valve apparatus to releasably secure the inlet end of the first one of the valve apparatus to the waste container at the container outlet of the waste container.

[0147] In some embodiments, the inlet end mating surface of the first one of the valve apparatus of the vacuum pump system engages with a corresponding container outlet mating surface of the waste container.

[0148] In some embodiments, the channel inlet seal of the first one of the valve apparatus of the vacuum pump system seals an interface between the inlet end mating surface of the first one of the

valve apparatus of the vacuum pump system and the container outlet mating surface of the waste container.

[0149] In some embodiments, the at least one wall of the waste container defines a container inlet for receiving the waste into the waste container; and the flow channel of the second one of the valve apparatus of the vacuum pump system is in fluid communication with the chamber of the waste container through the container inlet of the waste container and through the channel outlet of the second one of the valve apparatus of vacuum pump system.

[0150] In some embodiments, the outlet end of the second one of the valve apparatus of the vacuum pump system is connected to the waste container at the container inlet of the waste container.

[0151] In some embodiments, the channel outlet clamp of the second one of the valve apparatus of the vacuum pump system releasably secures the outlet end of the second one of the valve apparatus of the vacuum pump system to the waste container at the container inlet of the waste container.

[0152] In some embodiments, the waste container further comprises at least one waste container inlet interlocking portion at the container inlet, the at least one waste container inlet interlocking portion releasably interlocked with a corresponding one of the at least one outlet end interlocking portion of the outlet end of the second one of the valve apparatus to releasably secure the outlet end of the second one of the valve apparatus to the waste container at the container inlet of the waste container.

[0153] In some embodiments, the outlet end mating surface of the second one of the valve apparatus of the vacuum pump system engages with a corresponding container inlet mating surface of the waste container.

[0154] In some embodiments, the channel outlet seal of the second one of the valve apparatus of the vacuum pump system seals an interface between the outlet end mating surface of the second one of the valve apparatus of the vacuum pump system and the container inlet mating surface of the waste container.

[0155] In some embodiments, the waste container comprises the waste container disclosed above.

[0156] According to at least another embodiment, there is disclosed a kit comprising: the waste container disclosed above; and at least one of the valve apparatus disclosed above.

[0157] In some embodiments, the kit further comprises a pump.

[0158] In some embodiments, the pump comprises a vacuum pump.

[0159] In some embodiments, the at least one of the valve apparatus disclosed above comprises at least two of the valve apparatus disclosed above.

[0160] Other aspects and features will become apparent to those ordinarily skilled in the art upon review of the following description of illustrative embodiments in conjunction with the accompanying figures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0161] Exemplary embodiments are illustrated in referenced figures of the drawings. It is intended that the embodiments and figures disclosed herein are to be considered illustrative rather than restrictive.

[0162] FIG. 1 is a schematic plan view of a watercraft according to one embodiment.

[0163] FIG. 2 is a perspective view of a waste container according to one embodiment.

[0164] FIG. 3 is a side view of the waste container of FIG. 2.

[0165] FIG. 4 is a cutaway perspective view of the waste container of FIG. 2.

[0166] FIG. 5 is a perspective view of the waste container of FIG. 2 with an inlet pipe, outlet pipes, a top wall, and a dip tube of the waste container removed.

[0167] FIG. 6 is a cutaway perspective view of the waste container of FIG. 5, taken along the line labelled FIGS. 6, 7 in FIG. 5.

[0168] FIG. 7 is a cross-sectional view of the waste container of FIG. 5, taken along the line labelled FIGS. 6, 7 in FIG. 5.

[0169] FIG. 8 is a plan view of the waste container of FIG. 5.

[0170] FIG. 9 is a perspective view of a waste container and vacuum pump system according to another embodiment.

[0171] FIG. 10 is a cross-sectional view of the waste container of FIG. 9, taken along the line labelled FIG. 10 in FIG. 9.

[0172] FIG. 11 is an exploded view of the vacuum pump system of FIG. 9.

[0173] FIG. 12 is a perspective view of a valve apparatus of the vacuum pump system of FIG. 9.

[0174] FIG. 13 is a perspective view of the valve apparatus of FIG. 12, with a channel inlet clamp and a channel outlet clamp removed.

[0175] FIG. 14 is a cross-sectional view of the valve apparatus of FIG. 12, taken along the line labelled FIG. 14 in FIG. 12.

[0176] FIG. 15 is a perspective exploded view of a vacuum pump system according to another embodiment.

[0177] FIG. 16 is a front exploded view of the vacuum pump system of FIG. 15.

[0178] FIG. 17 is a front view of the vacuum pump system of FIG. 15 when assembled.

[0179] FIG. 18 is a perspective view of a valve apparatus of the vacuum pump system of FIG. 15.

[0180] FIG. 19 is a cross-sectional view of the valve apparatus of FIG. 18, taken along the line labelled FIG. 19 in FIG. 18.

DETAILED DESCRIPTION

[0181] Referring to FIG. 1, a vehicle waste management system according to one embodiment is shown generally at **100** and includes a watercraft **102** and a waste container **104**. In some embodiments, the watercraft **102** may be, for example, a wake boat, a water-ski boat, or another boat, aquatic vessel, or marine vessel. However, the watercraft **102** is an example only, and alternative embodiments may differ. For example, alternative embodiments may include a vehicle other than a watercraft, such as an aircraft or a recreational vehicle.

[0182] The watercraft **102** includes a hull **106** and an inboard engine **108** operable to generate an engine thrust force relative to the hull **106**. The engine thrust force generated by the inboard engine **108** may cause the watercraft **102** to move. The watercraft **102** may also move in response to external forces, such as wave action on the hull **106**. For example, the engine thrust force and/or the wave action may cause the watercraft **102** to move in a tilting motion. The inboard engine **108** is an example only, and alternative embodiments may differ. For example, alternative embodiments may include an inboard engine, a sterndrive engine, a jet-drive engine, a thruster engine, a surface-drive engine, a pod-drive engine, or any other engine or propulsion device, and alternative embodiments may include one, two, or more than two such engines or other propulsion devices.

[0183] In the embodiment shown, the waste container **104** is installed on the watercraft **102** within the hull **106** and is generally operable to receive and manage waste onboard the watercraft **102**. The waste container **104** may also be referred to as a “waste containment tank” or a “waste holding tank”. As used herein, the term “waste” may include, for example, liquid waste, solid waste, slurry, human waste, tissue, wastewater, used water, blackwater, and greywater. The waste container **104** may receive the waste, for example, from a toilet, sink, shower, or the like (not shown) onboard the watercraft **102**.

[0184] Referring now to FIGS. 2 to 8, a specific embodiment of the waste container **104** is shown and includes side walls **110**, **112**, **114**, and **116**, a top wall **117**, and a bottom wall **118**. In the embodiment shown, the top wall **117** includes a tower extension **119** extending from the top wall **117**. In some embodiments, the top wall **117**, including the tower extension **119**, may be removable from the side walls **110**, **112**, **114**, **116** and the bottom wall **118**. That is, the top wall **117** may

function as a cover or lid for the container **104**. Collectively, the side walls **110**, **112**, **114**, **116**, the top wall **117**, and the bottom wall **118** form an inner surface **120** of the waste container **104**. The inner surface **120** defines a chamber shown generally at **122**, which may hold waste produced onboard the watercraft **102**. The tower extension **119** of the top wall **117** defines a top inlet, shown generally at **124**, for receiving waste into the chamber **122**, and a top outlet, shown generally at **125**, for removing waste from the chamber **122**. The bottom wall **118** defines a bottom outlet, shown generally at **126**, for removing waste from the chamber **122**.

[0185] The top inlet **124** is generally configured to be in fluid communication with a toilet and/or a sink (not shown) onboard the watercraft **102** when the waste container **104** is installed on the watercraft **102** as shown in FIG. **1**. In the embodiment shown (see FIGS. **2** to **4**), the top inlet **124** is in fluid communication with the toilet and/or a sink via an inlet pipe **109**. The top outlet **125** is generally configured to be in fluid communication with a pump-out fitting (not shown) of the watercraft **102** when the waste container **104** is installed on the watercraft **102**. In the embodiment shown, the top outlet **125** is in fluid communication with the pump-out fitting via an outlet pipe **111**. The bottom outlet **126** is generally configured to be in fluid communication with a seacock (not shown) on the hull **106** of the watercraft **102** when the waste container **104** is installed on the watercraft **102**. In the embodiment shown, the bottom outlet **126** is in fluid communication with the seacock via an outlet pipe **113**. Thus, when the waste container **104** is installed on the watercraft **102**, the top inlet **124** may receive waste from the toilet and/or sink via the inlet pipe **109**, and waste may be removed from the waste container **104** through the top outlet **125** and out the pump-out fitting via the outlet pipe **111**, and/or through the bottom outlet **126** and out the seacock via the outlet pipe **113**. For example, waste may be removed through the top outlet **125** and out the pump-out fitting when the watercraft **102** is in port, while waste may be removed through the bottom outlet **126** and out the seacock when the watercraft **102** is in open waters.

[0186] In the embodiment shown, when the waste container **104** is installed on the watercraft **102** as shown in FIG. **1**, the waste container **104** is oriented such that the top wall **117** (including the tower extensions **119**) is generally a vertically uppermost, or top, boundary of the waste container **104**, and the bottom wall **118** is generally a vertically lowermost, or bottom, boundary of the waste container **104**—i.e., such that the top wall **117** generally faces a top of the watercraft **102** and the bottom wall **118** generally faces a bottom of the watercraft **102**. As such, in the embodiment shown, when the waste container **104** is installed on the watercraft **102** as shown in FIG. **1**, the top inlet **124** and the top outlet **125** are generally at a top **105** of the chamber **122**, and the bottom outlet **126** is generally at a bottom **107** of the chamber **122**. Due to the position of the bottom outlet **126** at the bottom **107** of the chamber **122**, emptying of waste from the chamber **122** through the bottom outlet **126** may be assisted by gravity. In the embodiment shown, the waste container **104** also includes a dip tube **115** extending from the top outlet **125** into the chamber **122**, terminating at the bottom **107** of the chamber **122** near the bottom wall **118**. The dip tube **115** is in fluid communication with both the top outlet **125** and the chamber **122**, and thus provides a conduit for waste to pass directly from the bottom **107** of the chamber **122** out the top outlet **125**, for example when vacuum suction is applied to the pump-out fitting.

[0187] In some embodiments, one or more of the top inlet **124**, the top outlet **125**, and the bottom outlet **126** may be reversibly closable. For example, in some embodiments, the waste container **104** may include one or more valves (not shown) operable to reversibly close the top inlet **124**, the top outlet **125**, and/or the bottom outlet **126**. The one or more valves may be, for example, ball valves.

[0188] The waste container **104** of the embodiment shown in FIGS. **2** to **8** is an example only, and alternative embodiments may differ. For example, alternative embodiments may include a waste container with a different shape and/or having a different number of walls. In some alternative embodiments, the waste container may have only a single wall (e.g., a generally spherically-shaped waste container). In other alternative embodiments, the waste container may include a plurality of walls where at least some of the walls are parallel with each other (e.g., a generally rectangular

cuboid-shaped waste container).

[0189] In the embodiment shown in FIGS. 2 to 8, the waste container **104** includes three openings in the chamber **122** for conveying waste into and out of the chamber **122**: the top inlet **124**, the top outlet **125**, and the bottom outlet **126**. However, in alternative embodiments, the waste container **104** may include only a single opening in the chamber **122**, or may include two openings or four or more openings in the chamber **122**. For example, in some alternative embodiments, the waste container **104** may include multiple inlets.

[0190] FIGS. 5 to 8 show the waste container **104** with the inlet pipe **109**, the outlet pipes **111** and **113**, the top wall **117**, and the dip tube **115** removed. Referring to FIGS. 5 to 8, the waste container **104** also includes projections **128**, **130**, and **132** projecting into the chamber **122** from the inner surface **120** at the side wall **110**; projections **134**, **136**, **138**, **140**, **142**, **144**, **146**, **148**, and **150** projecting into the chamber **122** from the inner surface **120** at the side wall **112**; projections **152**, **154**, and **156** projecting into the chamber **122** from the inner surface **120** at the side wall **114**; and projections **158**, **160**, **162**, **164**, **166**, **168**, **170**, **172**, and **174** projecting into the chamber **122** from the inner surface **120** at the side wall **116**. The projections **128**, **130**, **132**, **134**, **136**, **138**, **140**, **142**, **144**, **146**, **148**, **150**, **152**, **154**, **156**, **158**, **160**, **162**, **164**, **166**, **168**, **170**, **172**, and **174** may also be referred to as “paddles”, “ribs”, or “paddle ribs”.

[0191] In the embodiment shown, each of the projections **128**, **130**, **132**, **134**, **136**, **138**, **140**, **142**, **144**, **146**, **148**, **150**, **152**, **154**, **156**, **158**, **160**, **162**, **164**, **166**, **168**, **170**, **172**, and **174** projects into the chamber **122** from a respective portion of the inner surface **120** in a respective projection direction that is generally orthogonal to the respective portion of the inner surface **120**. For example, the projections **128**, **130**, and **132** project from a portion **176** of the inner surface **120** at the side wall **110** in a direction **178** that is generally orthogonal to the portion **176** of the inner surface **120**.

[0192] The projections **128**, **130**, **132**, **134**, **136**, **138**, **140**, **142**, **144**, **146**, **148**, **150**, **152**, **154**, **156**, **158**, **160**, **162**, **164**, **166**, **168**, **170**, **172**, and **174** may each project to varying depths into the chamber **122** from the inner surface **120**. In some embodiments, at least a portion of each of the projections **128**, **130**, **132**, **134**, **136**, **138**, **140**, **142**, **144**, **146**, **148**, **150**, **152**, **154**, **156**, **158**, **160**, **162**, **164**, **166**, **168**, **170**, **172**, and **174** may project about 0.5 inches, at least 0.5 inches, about 1.25 inches, or at least 1.25 inches into the chamber **122** from the inner surface **120**. In some embodiments, an entirety of each of the projections **128**, **130**, **132**, **134**, **136**, **138**, **140**, **142**, **144**, **146**, **148**, **150**, **152**, **154**, **156**, **158**, **160**, **162**, **164**, **166**, **168**, **170**, **172**, and **174** may project about 0.5 inches, at least 0.5 inches, about 1.25 inches, or at least 1.25 inches into the chamber **122** from the inner surface **120**. In some embodiments, each of the projections **128**, **130**, **132**, **134**, **136**, **138**, **140**, **142**, **144**, **146**, **148**, **150**, **152**, **154**, **156**, **158**, **160**, **162**, **164**, **166**, **168**, **170**, **172**, and **174** may project no more than 0.5 inches or 1.25 inches into the chamber **122** from the inner surface **120**.

[0193] In the embodiment shown, each of the projections **128**, **130**, **132**, **134**, **136**, **138**, **140**, **142**, **144**, **146**, **148**, **150**, **152**, **154**, **156**, **158**, **160**, **162**, **164**, **166**, **168**, **170**, **172**, and **174** extends along a portion of the inner surface **120** that is between the top inlet **124** and the bottom outlet **126**. More specifically, each of the projections **128**, **130**, **132**, **134**, **136**, **138**, **140**, **142**, **144**, **146**, **148**, **150**, **152**, **154**, **156**, **158**, **160**, **162**, **164**, **166**, **168**, **170**, **172**, and **174** generally extends in a direction from the top inlet **124** towards the bottom outlet **126**. Thus, when the waste container **104** is installed on the watercraft **102**, each of the projections **128**, **130**, **132**, **134**, **136**, **138**, **140**, **142**, **144**, **146**, **148**, **150**, **152**, **154**, **156**, **158**, **160**, **162**, **164**, **166**, **168**, **170**, **172**, and **174** extends along the inner surface **120** in a substantially vertical direction, generally from the top **105** to the bottom **107** of the chamber **122**.

[0194] In addition, in the embodiment shown, each of the projections **128**, **130**, **132**, **134**, **136**, **138**, **140**, **142**, **144**, **146**, **148**, **150**, **152**, **154**, **156**, **158**, **160**, **162**, **164**, **166**, **168**, **170**, **172**, and **174** extends along the inner surface **120** without contacting any other one of the projections **128**, **130**, **132**, **134**, **136**, **138**, **140**, **142**, **144**, **146**, **148**, **150**, **152**, **154**, **156**, **158**, **160**, **162**, **164**, **166**, **168**, **170**, **172**, and **174**. More specifically, in the embodiment shown, the projections on each of the side

walls **110**, **112**, **114**, and **116** are oriented parallel to one another and are spaced apart from one another at respective fixed projection spacings, each of the respective fixed projection spacings being in a spacing direction along the inner surface **120** perpendicular to a respective extension direction along the inner surface **120** of the corresponding projections. That is, the projections **128**, **130**, and **132** extend parallel to one another along the inner surface **120** at the side wall **110** and are spaced apart from each other at a fixed projection spacing **129** in a spacing direction **131** which is perpendicular to an extension direction **133** of the projections **128**, **130**, and **132**; the projections **134**, **136**, **138**, **140**, **142**, **144**, **146**, **148**, and **150** extend parallel to one another along the inner surface **120** at the side wall **112** and are spaced apart from each other at a fixed projection spacing **135** in a spacing direction **137** which is perpendicular to an extension direction **139** of the projections **134**, **136**, **138**, **140**, **142**, **144**, **146**, **148**, and **150**; the projections **152**, **154**, and **156** extend parallel to one another along the inner surface **120** at the side wall **114** and are spaced apart from each other at a fixed projection spacing **141** in a spacing direction **143** which is perpendicular to an extension direction **145** of the projections **152**, **154**, and **156**; and the projections **158**, **160**, **162**, **164**, **166**, **168**, **170**, **172**, and **174** extend parallel to one another along the inner surface **120** at the side wall **116** and are spaced apart from each other at a fixed projection spacing **147** in a spacing direction **149** which is perpendicular to an extension direction **151** of the projections **158**, **160**, **162**, **164**, **166**, **168**, **170**, **172**, and **174**. In the embodiment shown, the fixed projection spacings **129** and **141** are substantially equal to one another, and the fixed projection spacings **135** and **147** are substantially equal to one another. However, in some alternative embodiments, projections along each side wall may generally have the same fixed projection spacing, while in other alternative embodiments, projections along each side wall may have unique respective fixed projection spacings. Each of the fixed projection spacings **129**, **135**, **141**, and **147** may be, for example, at most 1 inch, about 1 inch, at least one inch, at most 2 inches, or about 2 inches.

[0195] More generally, the embodiment shown is an example only, and alternative embodiments may differ. For example, alternative embodiments may include more or fewer projections than the embodiment shown in FIGS. 2 to 8. In some alternative embodiments, the projections may project from the inner surface **120** in projection directions not necessarily orthogonal to the inner surface **120**. In some alternative embodiments, the projections may not extend in directions generally from the top inlet **124** to the bottom outlet **126**, but may instead extend in other directions. Moreover, in some alternative embodiments, projections along a given side wall of the waste container **104** may not all be parallel to each other, and in some embodiments, none of the projections along a given side wall may be parallel to any of the other projections along that side wall. Additionally, although in the embodiment shown in FIGS. 2 to 8 projections project from each of the side walls **110**, **112**, **114**, and **116** while no projections project from the top wall **117** or the bottom wall **118**, in alternative embodiments, projections may project from all walls of the waste container **104**, while in other alternative embodiments, projections may only project from a single wall of the waste container **104**, for example.

[0196] In the embodiment shown, the side walls **110**, **112**, **114**, and **116** of the waste container **104** also define a step-back **180** between the top wall **117** and the bottom wall **118**. The step-back **180** may provide structural support to the waste container **104**, and may also be used to help secure the waste container **104** when the waste container **104** is installed on the watercraft **102**, for example, by providing a location for securing a hold-down strap. In some embodiments, the projections **128**, **130**, **132**, **134**, **136**, **138**, **140**, **142**, **144**, **146**, **148**, **150**, **152**, **154**, **156**, **158**, **160**, **162**, **164**, **166**, **168**, **170**, **172**, and **174** may also provide structural support to the waste container **104**.

[0197] In operation, when the watercraft **102** moves, for example tilting due to engine thrust force generated by the inboard engine **108** and/or due to wave action on the hull **106**, such movement/tilting of the watercraft **102** may cause movement of the waste in the chamber **122**. As the waste moves in the chamber **122**, the projections **128**, **130**, **132**, **134**, **136**, **138**, **140**, **142**, **144**, **146**, **148**, **150**, **152**, **154**, **156**, **158**, **160**, **162**, **164**, **166**, **168**, **170**, **172**, and **174** may disturb or

obstruct flow of the waste in the chamber **122**. In some embodiments, the projections **128, 130, 132, 134, 136, 138, 140, 142, 144, 146, 148, 150, 152, 154, 156, 158, 160, 162, 164, 166, 168, 170, 172, and 174** may generate turbulence in the waste as the waste moves in the chamber **122**. The turbulence thus generated may help break apart solids in the waste and disperse particles in the waste. In this way, the projections **128, 130, 132, 134, 136, 138, 140, 142, 144, 146, 148, 150, 152, 154, 156, 158, 160, 162, 164, 166, 168, 170, 172, and 174** may accelerate waste break-down in the waste container **104**, and may thus enhance waste management onboard the watercraft **102**.

[0198] Generally, the waste container **104** may be manufactured by molding the side walls **110, 112, 114, 116**, the top wall **117**, and the bottom wall **118** from resin. For example, the side walls **110, 112, 114, 116**, the top wall **117**, and the bottom wall **118** may be molded using a rotational molding process and/or an injection molding process. The side walls **110, 112, 114, 116** and the bottom wall **118** may be molded separately from the top wall **117** (for example, in embodiments where the top wall **117** is removable). In some embodiments, the projections **128, 130, 132, 134, 136, 138, 140, 142, 144, 146, 148, 150, 152, 154, 156, 158, 160, 162, 164, 166, 168, 170, 172, and 174** may be molded in with the side walls **110, 112, 114, 116**. In other embodiments, the projections **128, 130, 132, 134, 136, 138, 140, 142, 144, 146, 148, 150, 152, 154, 156, 158, 160, 162, 164, 166, 168, 170, 172, and 174** may be fixed or attached to the inner surface **120** at the side walls **110, 112, 114, 116** after the side walls **110, 112, 114, 116** have been formed.

[0199] Referring now to FIGS. **9** and **10**, an alternative embodiment which may be used, for example, with the vehicle waste management system **100** is shown and includes a waste container **182** and a vacuum pump system **183**. The waste container **182** may be similar to the waste container **104** of the embodiment of FIGS. **2** to **8**, and may be used instead of the waste container **104** onboard the watercraft **102**. The waste container **182** includes side walls **184, 186, 188, and 190**, a top wall **192**, and a bottom wall **194**. In some embodiments, the top wall **192** may be removable from the side walls **184, 186, 188, and 190** and the bottom wall **194**. That is, the top wall **192** may function as a cover or lid for the container **182**. Collectively, the side walls **184, 186, 188, and 190**, the top wall **192**, and the bottom wall **194** form an inner surface **196** of the waste container **182**. The inner surface **196** defines a main chamber shown generally at **198**, which may hold waste produced onboard the watercraft **102**. Inside the main chamber **198**, below the top wall **192**, the waste container **182** includes a vacuum reservoir **200** defining a separate vacuum chamber shown generally at **202**. The vacuum reservoir **200** and the top wall **192** of the waste container **182** define a top vacuum chamber inlet, shown generally at **204**, for receiving waste into the vacuum chamber **202**, and a top vacuum chamber outlet, shown generally at **206**, for removing waste from the vacuum chamber **202**. The top wall **192** also defines a top inlet shown generally at **208**, for receiving waste into the chamber **198**, and a top outlet, shown generally at **210**, for removing waste from the chamber **198**. Additionally, the bottom wall **194** defines a bottom outlet, shown generally at **212**, for removing waste from the main chamber **198**.

[0200] The top vacuum chamber inlet **204** is generally configured to be in fluid communication with a toilet and/or a sink (not shown) onboard the watercraft **102** when the waste container **182** is installed on the watercraft **102** as shown in FIG. **1**. In the embodiment shown, the top vacuum chamber inlet **204** is in fluid communication with the toilet and/or a sink via an inlet pipe **214**. The top vacuum chamber outlet **206** of the vacuum chamber **202** and the top inlet **208** of the main chamber **198** of the waste container **182** are generally configured to be in fluid communication with the vacuum pump system **183**. In the embodiment shown, the top vacuum chamber outlet **206** is in fluid communication with the vacuum pump system **183** via a pump system inlet fitting **216** of the waste container **182**, and the top inlet **208** is in fluid communication with the vacuum pump system **183** via a pump system outlet fitting **218** of the waste container **182**. The top outlet **210** of the main chamber **198** of the waste container **182** is generally configured to be in fluid communication with a pump-out fitting (not shown) of the watercraft **102** when the waste container **182** is installed on the watercraft **102**. In the embodiment shown, the top outlet **210** is in fluid communication with the

pump-out fitting via an outlet pipe **220**. The bottom outlet **212** is generally configured to be in fluid communication with a seacock (not shown) on the hull **106** of the watercraft **102** when the waste container **182** is installed on the watercraft **102**. In the embodiment shown, the bottom outlet **212** is in fluid communication with the seacock via an outlet pipe **222**. Thus, when the waste container **182** and the vacuum pump system **183** are installed on the watercraft **102**, the top vacuum chamber inlet **204** may receive waste, via the inlet pipe **214**, from the toilet and/or sink into the vacuum chamber **202**, the waste may be transferred from the vacuum chamber **202** to the main chamber **198** out the through the top vacuum chamber outlet **206**, the pump system inlet fitting **216**, the vacuum pump system **183**, the pump system outlet fitting **218**, and the top inlet **208**, and the waste may be removed from the main chamber **198** through the top outlet **210** and out the pump-out fitting via the outlet pipe **220**, and/or through the bottom outlet **212** and out the seacock via the outlet pipe **222**. For example, waste may be removed through the top outlet **210** and out the pump-out fitting when the watercraft **102** is in port, while waste may be removed through the bottom outlet **212** and out the seacock when the watercraft **102** is in open waters.

[0201] In the embodiment shown, when the waste container **182** is installed on the watercraft **102** as shown in FIG. 1, the waste container **182** is oriented such that the top wall is generally a vertically uppermost, or top, boundary of the waste container **182**, and the bottom wall **194** is generally a vertically lowermost, or bottom, boundary of the waste container **182**—i.e., such that the top wall **192** generally faces a top of the watercraft **102** and the bottom wall **194** generally faces a bottom of the watercraft **102**. As such, in the embodiment shown, when the waste container **182** is installed on the watercraft **102** as shown in FIG. 1, the top vacuum chamber inlet **204** and the top outlet **210** are generally at a top **224** of the main chamber **198** (and of the vacuum chamber **202**), and the bottom outlet **212** is generally at a bottom **226** of the main chamber **198**. Due to the position of the bottom outlet **212** at the bottom **226** of the main chamber **198**, emptying of waste from the main chamber **198** through the bottom outlet **212** may be assisted by gravity. In the embodiment shown, the waste container **182** also includes a dip tube **225** extending from the top outlet **210** into the chamber **182**, terminating at the bottom **226** of the chamber **182** near the bottom wall **194**. The dip tube **225** is in fluid communication with both the top outlet **210** and the chamber **182**, and thus provides a conduit for waste to pass directly from the bottom **226** of the chamber **182** out the top outlet **210**, for example when vacuum suction is applied to the pump-out fitting.

[0202] In some embodiments, one or more of the top vacuum chamber inlet **204**, the top outlet **210**, and the bottom outlet **212** may be reversibly closable. For example, in some embodiments, the waste container **182** may include one or more valves (not shown) operable to reversibly close the top vacuum chamber inlet **204**, the top outlet **210**, and/or the bottom outlet **212**. The one or more valves may be, for example, ball valves.

[0203] The waste container **182** of the embodiment shown in FIGS. 9 and 10 is an example only, and alternative embodiments may differ. For example, alternative embodiments may include a waste container with a different shape and/or having a different number of walls. In some alternative embodiments, the waste container may have only a single wall (e.g., a generally spherically-shaped waste container). In other alternative embodiments, the waste container may include a plurality of walls where at least some of the walls are parallel with each other (e.g., a generally rectangular cuboid-shaped waste container).

[0204] In the embodiment shown in FIGS. 9 and 10, the waste container **182** includes five openings: two in the vacuum chamber **202** for conveying waste into and out of the vacuum chamber **202**—the top vacuum chamber inlet **204** and the top vacuum chamber outlet **206**; and three in the main chamber **198** for conveying waste into and out of the main chamber **198**—the top inlet **208**, the top outlet **210**, and the bottom outlet **212**. However, in alternative embodiments, the waste container **182** may include different numbers of openings in the vacuum chamber **202** and/or in the main chamber **198**. For example, in some alternative embodiments, the waste container **182** may include multiple vacuum chamber inlets.

[0205] Referring now to FIGS. 9, 10, and 11, the vacuum pump system 183 includes a vacuum pump 228, an upstream valve apparatus 230, and a downstream valve apparatus 232. Generally, the vacuum pump system 183 is configured to control a flow of waste and/or other fluids from the vacuum chamber 202 to the main chamber 198. More specifically, the vacuum pump 228 is operable to cause the waste and/or other fluids to flow through the pump system 183, while the upstream valve apparatus 230 and the downstream valve apparatus 232 control the direction of flow. The vacuum pump 228 includes a pump inlet, shown generally at 234 (see FIG. 10), for receiving the waste into the vacuum pump 228, and a pump outlet, shown generally at 236, for removing the waste from the vacuum pump 228. Of course, the vacuum pump system 183 is an example only, and alternative embodiments may differ. For example, some alternative embodiments may employ pumps other than vacuum pumps.

[0206] The upstream valve apparatus 230 is generally similar to the downstream valve apparatus 232, and each of the valve apparatuses 230 and 232 may generally be configured for use with a pump, such as the vacuum pump 228, to control a flow of waste into or out of a waste container, such as the waste container 182 or the vacuum reservoir 200. The upstream valve apparatus 230 includes a valve body 238 having an inlet end 240 and an outlet end 242. The inlet end 240 defines a channel inlet, shown generally at 244, and the outlet end 242 defines a channel outlet, shown generally at 246. Between the channel inlet 244 and the channel outlet 246, the valve body 238 defines a flow channel, shown generally at 248, for conveying waste. That is, the flow channel 248 includes and extends between the channel inlet 244 and the channel outlet 246. Similarly, the downstream valve apparatus 232 includes a valve body 250 having an inlet end 252 and an outlet end 254. The inlet end 252 defines a channel inlet, shown generally at 256, and the outlet end 254 defines a channel outlet, shown generally at 258. Between the channel inlet 256 and the channel outlet 258, the valve body 250 defines a flow channel, shown generally at 260, for conveying waste. That is, the flow channel 260 includes and extends between the channel inlet 256 and the channel outlet 258.

[0207] Generally, each of the inlet ends 240 and 252 is connectable to a waste source to cause the respective flow channel 248 or 260 to be in fluid communication with the waste source through the respective channel inlet 244 or 256, such that the respective channel inlet 244 or 256 may receive waste into the respective flow channel 248 or 260. Similarly, each of the outlet ends 242 and 254 is connectable to a waste destination to cause the respective flow channel 248 or 260 to be in fluid communication with the waste destination through the respective channel outlet 246 or 258, such that the respective channel outlet 246 or 258 may remove waste from the respective flow channel 248 or 260. The waste source may be, for example, a waste container such as the waste container 104, the waste container 182, or the vacuum reservoir 200; or a pump such as the vacuum pump 228. Similarly, the waste destination may be, for example, a waste container such as the waste container 104, the waste container 182, or the vacuum reservoir 200; or a pump such as the vacuum pump 228.

[0208] In the embodiment shown, the inlet end 240 of the upstream valve apparatus 230 is connected to the pump system inlet fitting 216 and thus the vacuum reservoir 200 and, more generally, the waste container 182, causing the flow channel 248 to be in fluid communication with the vacuum reservoir 200 (i.e., the waste source of the upstream valve apparatus 230) through the channel inlet 244. The outlet end 242 of the upstream valve apparatus 230 is connected to the pump inlet 234, causing the flow channel 248 to be in fluid communication with the vacuum pump 228 (i.e., the waste destination of the upstream valve apparatus 230) through the channel outlet 246. The inlet end 252 of the downstream valve apparatus 232 is connected to the pump outlet 236, causing the flow channel 260 to be in fluid communication with the vacuum pump 228 (i.e., the waste source of the downstream valve apparatus 232) through the channel inlet 256. The outlet end 254 of the downstream valve apparatus 232 is connected to the pump system outlet fitting 218 and thus the waste container 182, causing the flow channel 260 to be in fluid communication with the

waste container **182** (i.e., the waste destination of the downstream valve apparatus **232**) through the channel outlet **258**. However, the embodiment shown is an example only, and alternative embodiments may differ. For example, in some alternative embodiments, the inlet end **240** of the upstream valve apparatus **230** and/or the outlet end **254** of the downstream valve apparatus **232** may be connected directly to the waste container **182** (i.e., as opposed to connecting through the pump system inlet fitting **216** or the pump system outlet fitting **218**).

[0209] Each of the valve apparatuses **230** and **232** also includes a pair of clamps for releasably securing the valve apparatuses **230** and **232** to their respective waste sources and/or waste destinations. More specifically, the upstream valve apparatus **230** includes a channel inlet clamp **262** operable to releasably secure the inlet end **240** of the upstream valve apparatus **230** to a waste source of the upstream valve apparatus **230** to connect the inlet end **240** to the waste source to cause the flow channel **248** to be in fluid communication with the waste source through the channel inlet **244**, and a channel outlet clamp **264** operable to releasably secure the outlet end **242** of the upstream valve apparatus **230** to a waste destination of the upstream valve apparatus **230** to connect the outlet end **242** to the waste destination to cause the flow channel **248** to be in fluid communication with the waste destination through the channel outlet **246**. Similarly, the downstream valve apparatus **232** includes a channel inlet clamp **266** operable to releasably secure the inlet end **252** of the downstream valve apparatus **232** to a waste source of the downstream valve apparatus **232** to connect the inlet end **252** to the waste source to cause the flow channel **260** to be in fluid communication with the waste source through the channel inlet **256**, and a channel outlet clamp **268** operable to releasably secure the outlet end **254** of the downstream valve apparatus **232** to a waste destination of the downstream valve apparatus **232** to connect the outlet end **254** to the waste destination to cause the flow channel **260** to be in fluid communication with the waste destination through the channel outlet **258**. In the embodiment shown, the channel inlet clamp **262** releasably secures and connects the inlet end **240** of the upstream valve apparatus **230** to the pump system inlet fitting **216**, the channel outlet clamp **264** releasably secures and connects the outlet end **242** of the upstream valve apparatus **230** to the pump inlet **234**, the channel inlet clamp **266** releasably secures and connects the inlet end **252** of the downstream valve apparatus **232** to the pump outlet **236**, and the channel outlet clamp **268** releasably secures and connects the outlet end **254** of the downstream valve apparatus **232** to the pump system outlet fitting **218**. Thus, the upstream valve apparatus **230** is connected and releasably secured to the pump system inlet fitting **216** and to the vacuum pump **228**, and the downstream valve apparatus **232** is connected and releasably secured to the vacuum pump **228** and to the pump system outlet fitting **218**.

[0210] FIGS. **12**, **13**, and **14** show the upstream valve apparatus **230** in greater detail. However, the following explanation and references to FIGS. **12**, **13**, and **14** are also applicable to the downstream valve apparatus **232**, which, as noted above, is generally similar to the upstream valve apparatus **230**. Referring to FIGS. **12**, **13**, and **14**, the inlet end **240** of the upstream valve apparatus **230** includes an inlet end mating surface **270** around the channel inlet **244**, and the outlet end **242** of the upstream valve apparatus **230** includes an outlet end mating surface **272** around the channel outlet **246**. Generally, the inlet end mating surface **270** is configured to engage with a corresponding waste source mating surface of a waste source of the upstream valve apparatus **230** when the inlet end **240** is connected to the waste source. Similarly, the outlet end mating surface **272** is configured to engage with a corresponding waste destination mating surface of a waste destination of the upstream valve apparatus **230** when the outlet end **242** is connected to the waste destination. In the embodiment shown, the inlet end mating surface **270** engages with a corresponding mating surface (not shown) of the pump system inlet fitting **216**, and the outlet end mating surface **272** engages with a corresponding pump inlet mating surface (not shown) of the vacuum pump **228**. However, alternative embodiments may differ. For example, in some alternative embodiments, the inlet end mating surface **270** may engage with a corresponding mating surface that is directly on the top wall **192** of the waste container **182**.

[0211] The upstream valve apparatus **230** also includes a channel inlet seal **274** seated on the inlet end mating surface **270** around the channel inlet **244**, and a channel outlet seal **276** seated on the outlet end mating surface **272** around the channel outlet **246**. Generally, the channel inlet seal **274** is configured to seal an interface between the inlet end mating surface **270** and a corresponding waste source mating surface of a waste source of the upstream valve apparatus **230** when the inlet end mating surface **270** engages with the waste source mating surface, and the channel outlet seal **276** is configured to seal an interface between the outlet end mating surface **272** and a corresponding waste destination mating surface of a waste destination of the upstream valve apparatus **230** when the outlet end mating surface **272** engages with the waste destination mating surface. In the embodiment shown, the channel inlet seal **274** seals an interface between the inlet end mating surface **270** and the corresponding mating surface of the pump system inlet fitting **216**, and the channel outlet seal **276** seals an interface between the outlet end mating surface **272** and the corresponding pump inlet mating surface of the vacuum pump **228**. In the embodiment shown, each of the channel inlet seal **274** and the channel outlet seal **276** includes a face seal and an O-ring. However, the embodiment shown is an example only, and alternative embodiments may differ. For example, some alternative embodiments may include inlet/outlet seals which include only a face seal, or which include only an O-ring, or which include other types of mechanical seal.

[0212] The upstream valve apparatus **230** also includes a first check valve **278** and a second check valve **280**. The first check valve **278** and the second check valve **280** are retained and secured within the flow channel **248** by the valve body **238**. Each of the first check valve **278** and the second check valve **280** is configured to allow waste to flow through the flow channel **248** in a forward flow direction **282** from the channel inlet **244** to the channel outlet **246**, and to prevent the waste from flowing through the flow channel **248** in a reverse flow direction **284** from the channel outlet **246** to the channel inlet **244**. Thus, the upstream valve apparatus **230** is generally configured to allow waste to flow from the channel inlet **244** to the channel outlet **246** (in the forward flow direction **282**), and to prevent waste from flowing from the channel outlet **246** to the channel inlet **244** (in the reverse flow direction **284**). Similarly, the downstream valve apparatus **232** is generally configured to allow waste to flow from the channel inlet **252** to the channel outlet **254**, and to prevent waste from flowing from the channel outlet **254** to the channel inlet **252**. Consequently, when the upstream valve apparatus **230** and the downstream valve apparatus **232** are connected and releasably secured to the vacuum pump **228** as shown in FIGS. **9** and **10**, such that the flow channel **248** of the upstream valve apparatus **230** is in fluid communication with the vacuum pump **228** through the pump inlet **234** and the flow channel **260** of the downstream valve apparatus **232** is in fluid communication with the vacuum pump **228** through the pump outlet **236**, the upstream valve apparatus **230** generally allows waste to flow into the vacuum pump **228** through the pump inlet **234** while preventing waste from flowing out of the vacuum pump **228** through the pump inlet **234**, and the downstream valve apparatus **232** generally allows waste to flow out of the vacuum pump **228** through the pump outlet **236** while preventing waste from flowing into the vacuum pump **228** through the pump outlet **236**. Therefore, the upstream valve apparatus **230** and the downstream valve apparatus **232** effectively control the direction of flow of waste through the vacuum pump **228** when the vacuum pump **228** causes the waste to flow through the vacuum pump system **183**, generally allowing the waste to flow in a direction from the pump inlet **234** to the pump outlet **236**, and generally preventing the waste from flowing in a direction from the pump outlet **236** to the pump inlet **234**. Thus, overall, when the vacuum pump **228** causes waste to flow through the vacuum pump system **183**, the waste may flow through the vacuum pump system **183** by entering the upstream valve apparatus **230** (and thus the vacuum pump system **183**) through the channel inlet **244**, passing through the flow channel **248**, leaving the upstream valve apparatus **230** through the channel outlet **246**, entering the vacuum pump **228** through the pump inlet **234**, passing through the vacuum pump **228**, leaving the vacuum pump **228** through the pump outlet **236**, entering the downstream valve apparatus **232** through the channel inlet **256**, passing through the flow channel

260, and leaving the downstream valve apparatus **232** (and thus the vacuum pump system **183**) through the channel outlet **258**.

[0213] In the embodiment shown, each of the first check valve **278** and the second check valve **280** of the upstream valve apparatus **230** is a duckbill valve, and a portion **286** of the first check valve **278** is received in the second check valve **280**. That is, the first check valve **278** is nested within second check valve **280**. However, the embodiment shown is an example only, and alternative embodiments may differ. For example, some alternative embodiments may include other types of check valves instead of or in addition to duckbill valves. Additionally, in some alternative embodiments, the valve apparatus may include a different number of check valves in the flow channel, such as only a single check valve in the flow channel, or three or more check valves in the flow channel. Some alternative embodiments may include check valves which are not nested (see, for example, FIG. **19**).

[0214] In the embodiment shown, the valve body **238** of the upstream valve apparatus **230** includes valve body portions **288**, **290**, and **292** arranged between the channel inlet **244** and the channel outlet **246**. The valve body portions **288**, **290**, and **292** collectively define the flow channel **248** and retain the first and second check valves **278** and **280** in the flow channel **248**. The valve body portion **288** is non-dismantlably fixed to the valve body portion **290**, which is in turn non-dismantlably fixed to the valve body portion **292**. That is, each of the valve body portions **288**, **290**, and **292** is non-dismantlably fixed to at least one other of the valve body portions **288**, **290**, and **292**, such that the valve body **238** as a whole is non-dismantlable. The valve body portions **288**, **290**, and **292** may be fixed together by, for example, plastic welding and/or glue. Of course, the embodiment shown is an example only, and alternative embodiments may differ. For example, in alternative embodiments, the valve body **238** may include a different number of valve body portions non-dismantlably fixed together.

[0215] Generally, the upstream valve apparatus **230**, or a similar valve apparatus such as the downstream valve apparatus **232**, may be installed on to a waste source (such as the pump system inlet fitting **216** of the waste container **182**) by positioning the inlet end **240** at the waste source and causing the channel inlet clamp **262** to secure the inlet end **240** to the waste source to connect the inlet end **240** to the waste source to cause the flow channel **248** to be in fluid communication with the waste source through the channel inlet **244**. The upstream valve apparatus **230** may then be uninstalled from the waste source by causing the channel inlet clamp **262** to release the inlet end **240** from the waste source. Similarly, the upstream valve apparatus **230** may be installed on to a waste destination (such as the pump inlet **234** of the vacuum pump **228**) by positioning the outlet end **242** at the waste destination and causing the channel outlet clamp **264** to secure the outlet end **242** to the waste destination to connect the outlet end **242** to the waste destination to cause the flow channel **248** to be in fluid communication with the waste destination through the channel outlet **246**. The upstream valve apparatus **230** may then be uninstalled from the waste destination by causing the channel outlet clamp **264** to release the outlet end **242** from the waste destination. That is, installation and uninstallation of the upstream valve apparatus **230**, or of a similar valve apparatus such as the downstream valve apparatus **232**, require only the use/operation of the channel inlet clamp **262** and/or the channel outlet clamp **264**. As such, valve apparatuses such as the upstream valve apparatus **230** and the downstream valve apparatus **232** disclosed herein may simplify maintenance (e.g., valve replacement) of waste management systems such as the waste management system **100**.

[0216] Referring now to FIGS. **15**, **16**, and **17**, an alternative embodiment of a vacuum pump system is shown generally at **294** and includes a vacuum pump **296**, an upstream valve apparatus **298**, and a downstream valve apparatus **300**. The vacuum pump system **294** may be used, for example, with the waste container **182** in the vehicle waste management system **100**, instead of the vacuum pump system **183**. That is, in embodiments where the vehicle waste management system **100** includes the vacuum pump system **294** installed with the waste container **182**, the top vacuum

chamber outlet **206** of the vacuum chamber **202** of the waste container **182** may be in fluid communication with the vacuum pump system **294** via the pump system inlet fitting **216**, and the top inlet **208** of the main chamber **198** of the waste container **182** may be in fluid communication with the vacuum pump system **294** via the pump system outlet fitting **218** (see also FIGS. **9** and **10**). Thus, when the waste container **182** and the vacuum pump system **294** are installed on the watercraft **102**, the top vacuum chamber inlet **204** may receive waste, via the inlet pipe **214**, from the toilet and/or sink into the vacuum chamber **202**, the waste may be transferred from the vacuum chamber **202** to the main chamber **198** out the through the top vacuum chamber outlet **206**, the pump system inlet fitting **216**, the vacuum pump system **294**, the pump system outlet fitting **218**, and the top inlet **208**, and the waste may be removed from the main chamber **198** through the top outlet **210** and out the pump-out fitting via the outlet pipe **220**, and/or through the bottom outlet **212** and out the seacock via the outlet pipe **222**. In such embodiments, the vacuum pump system **294**, much like the vacuum pump system **183** of the embodiment of FIGS. **9**, **10**, and **11**, may generally be configured to control a flow of waste and/or other fluids from the vacuum chamber **202** to the main chamber **198**. More specifically, the vacuum pump **296** is operable to cause the waste and/or other fluids to flow through the pump system **294**, while the upstream valve apparatus **298** and the downstream valve apparatus **300** control the direction of flow. The vacuum pump **296** may generally be similar to the vacuum pump **228**, and includes a pump inlet, shown generally at **302**, for receiving the waste into the vacuum pump **296**, and a pump outlet, shown generally at **304**, for removing the waste from the vacuum pump **296**.

[0217] The upstream valve apparatus **298** is generally similar to the downstream valve apparatus **300**, and each of the valve apparatuses **298** and **300** may generally be configured for use with a pump, such as the vacuum pump **296**, to control a flow of waste into or out of a waste container, such as the waste container **182** or the vacuum reservoir **200**. The upstream valve apparatus **298** includes a valve body **306** having an inlet end **308** and an outlet end **310**. The inlet end **308** defines a channel inlet, shown generally at **312**, and the outlet end **310** defines a channel outlet, shown generally at **314**. Between the channel inlet **312** and the channel outlet **314**, the valve body **306** defines a flow channel, shown generally at **316**, for conveying waste. That is, the flow channel **316** includes and extends between the channel inlet **312** and the channel outlet **314**. Similarly, the downstream valve apparatus **300** includes a valve body **318** having an inlet end **320** and an outlet end **322**. The inlet end **320** defines a channel inlet, shown generally at **324**, and the outlet end **322** defines a channel outlet, shown generally at **326**. Between the channel inlet **324** and the channel outlet **326**, the valve body **318** defines a flow channel, shown generally at **328**, for conveying waste. That is, the flow channel **328** includes and extends between the channel inlet **324** and the channel outlet **326**.

[0218] Generally, each of the inlet ends **308** and **320** is connectable to a waste source to cause the respective flow channel **316** or **328** to be in fluid communication with the waste source through the respective channel inlet **312** or **324**, such that the respective channel inlet **312** or **324** may receive waste into the respective flow channel **316** or **328**. Similarly, each of the outlet ends **310** and **322** is connectable to a waste destination to cause the respective flow channel **316** or **328** to be in fluid communication with the waste destination through the respective channel outlet **314** or **326**, such that the respective channel outlet **314** or **326** may remove waste from the respective flow channel **316** or **328**. The waste source may be, for example, a waste container such as the waste container **104**, the waste container **182**, or the vacuum reservoir **200**; or a pump such as the vacuum pump **296**. Similarly, the waste destination may be, for example, a waste container such as the waste container **104**, the waste container **182**, or the vacuum reservoir **200**; or a pump such as the vacuum pump **296**.

[0219] In some embodiments, the inlet end **308** of the upstream valve apparatus **298** may be connected to the pump system inlet fitting **216** and thus the vacuum reservoir **200** and, more generally, the waste container **182**, causing the flow channel **316** to be in fluid communication with

the vacuum reservoir **200** (i.e., the waste source of the upstream valve apparatus **298**) through the channel inlet **312**. The outlet end **310** of the upstream valve apparatus **298** is connected to the pump inlet **302**, causing the flow channel **316** to be in fluid communication with the vacuum pump **296** (i.e., the waste destination of the upstream valve apparatus **298**) through the channel outlet **314**. The inlet end **320** of the downstream valve apparatus **300** is connected to the pump outlet **304**, causing the flow channel **328** to be in fluid communication with the vacuum pump **296** (i.e., the waste source of the downstream valve apparatus **300**) through the channel inlet **324**. The outlet end **322** of the downstream valve apparatus **300** may be connected to the pump system outlet fitting **218** and thus the waste container **182**, causing the flow channel **328** to be in fluid communication with the waste container **182** (i.e., the waste destination of the downstream valve apparatus **300**) through the channel outlet **326**. In some alternative embodiments, the inlet end **308** of the upstream valve apparatus **298** and/or the outlet end **322** of the downstream valve apparatus **300** may instead be connected directly to the waste container **182** (i.e., as opposed to connecting through the pump system inlet fitting **216** or the pump system outlet fitting **218**).

[0220] Each of the valve apparatus **298** and the valve apparatus **300** also includes a pair of inlet end interlocking portions at its inlet end (i.e., **308** and **320**, respectively) and a pair of outlet end interlocking portions at its outlet end (i.e., **310** and **322**, respectively). In general, the inlet end interlocking portions may be used to releasably secure the valve apparatuses **298** and **300** to their respective waste sources, while the outlet end interlocking portions may be used to releasably secure the valve apparatuses **298** and **300** to their respective waste destinations.

[0221] More specifically, the upstream valve apparatus **298** includes inlet end interlocking portions **330** and **332** at the inlet end **308**, and outlet end interlocking portions **334** and **336** at the outlet end **310**. In the embodiment shown, the inlet end interlocking portions **330** and **332** project out from the inlet end **308** of the valve body **306** away from the flow channel **316**, and the outlet end interlocking portions **334** and **336** project out from the outlet end **310** of the valve body **306** away from the flow channel **316**. The inlet end interlocking portions **330** and **332** are releasably interlockable with corresponding waste source interlocking portions of a waste source to releasably secure the inlet end **308** to the waste source to connect the inlet end **308** to the waste source to cause the flow channel **316** to be in fluid communication with the waste source through the channel inlet **312**. The outlet end interlocking portions **334** and **336** are releasably interlockable with corresponding waste destination interlocking portions of a waste destination to releasably secure the outlet end **310** to the waste destination to connect the outlet end **310** to the waste destination to cause the flow channel **316** to be in fluid communication with the waste destination through the channel outlet **314**. In the embodiment shown, the inlet end interlocking portions **330** and **332** are releasably interlockable with the corresponding waste source interlocking portions by rotation of the upstream valve apparatus **298**, relative to the waste source, around an inlet channel axis **338** which is parallel to the flow channel **316** at the channel inlet **312**, and the outlet end interlocking portions **334** and **336** are releasably interlockable with the corresponding waste destination interlocking portions by rotation of the upstream valve apparatus **298**, relative to the waste source, around an outlet channel axis **340** which is parallel to the flow channel **316** at the channel outlet **314**. For example, as shown in FIGS. **16** and **17**, the vacuum pump **296**—i.e., the waste destination of the upstream valve apparatus **298**—includes vacuum pump inlet interlocking portions **342** and **344** projecting out from the vacuum pump **296** at the pump inlet **302**. The vacuum pump inlet interlocking portions **342** and **344** are sized and shaped to slidably receive and releasably interlock with the outlet end interlocking portions **334** and **336** of the upstream valve apparatus **298** when the upstream valve apparatus **298** and the vacuum pump **296** are rotated relative to each other around the outlet channel axis **340**. Thus, the outlet end **310** of the upstream valve apparatus **298** may be releasably secured and connected to the vacuum pump **296** at the pump inlet **302** by positioning the outlet end **310** at the pump inlet **302** and rotating the upstream valve apparatus **298**, relative to the to the vacuum pump **296**, around the outlet channel axis **340** to cause the outlet end interlocking

portions **334** and **336** to slide against and releasably interlock with the vacuum pump inlet interlocking portions **342** and **344**, respectively, as shown in FIG. **17**. This securing mechanism based on rotational interlocking of corresponding projections on a valve apparatus and a waste source/destination may also be referred to as a “twist lock” securing mechanism.

[0222] Similarly, the downstream valve apparatus **300** includes inlet end interlocking portions **346** and **348** at the inlet end **320**, and outlet end interlocking portions **350** and **352** at the outlet end **322**. In the embodiment shown, the inlet end interlocking portions **346** and **348** project out from the inlet end **320** of the valve body **318** away from the flow channel **328**, and the outlet end interlocking portions **350** and **352** project out from the outlet end **322** of the valve body **318** away from the flow channel **328**. The inlet end interlocking portions **346** and **348** are releasably interlockable with corresponding waste source interlocking portions of a waste source to releasably secure the inlet end **320** to the waste source to connect the inlet end **320** to the waste source to cause the flow channel **328** to be in fluid communication with the waste source through the channel inlet **324**. The outlet end interlocking portions **350** and **352** are releasably interlockable with corresponding waste destination interlocking portions of a waste destination to releasably secure the outlet end **322** to the waste destination to connect the outlet end **322** to the waste destination to cause the flow channel **328** to be in fluid communication with the waste destination through the channel outlet **326**. In the embodiment shown, the inlet end interlocking portions **346** and **348** are releasably interlockable with the corresponding waste source interlocking portions by rotation of the downstream valve apparatus **300**, relative to the waste source, around an inlet channel axis **354** which is parallel to the flow channel **328** at the channel inlet **324**, and the outlet end interlocking portions **350** and **352** are releasably interlockable with the corresponding waste destination interlocking portions by rotation of the downstream valve apparatus **300**, relative to the waste source, around an outlet channel axis **356** which is parallel to the flow channel **328** at the channel outlet **326**. For example, as shown in FIGS. **16** and **17**, the vacuum pump **296**—i.e., the waste source of the downstream valve apparatus **300**—includes vacuum pump outlet interlocking portions **358** and **360** projecting out from the vacuum pump **296** at the pump outlet **304**. The vacuum pump outlet interlocking portions **358** and **360** are sized and shaped to slidably receive and releasably interlock with the inlet end interlocking portions **346** and **348** of the downstream valve apparatus **300** when the downstream valve apparatus **300** and the vacuum pump **296** are rotated relative to each other around the inlet channel axis **354**. Thus, the inlet end **320** of the downstream valve apparatus **300** may be releasably secured and connected to the vacuum pump **296** at the pump outlet **304** by positioning the inlet end **320** at the pump outlet **304** and rotating the downstream valve apparatus **300**, relative to the vacuum pump **296**, around the inlet channel axis **354** to cause the inlet end interlocking portions **346** and **348** to slide against and releasably interlock with the vacuum pump outlet interlocking portions **358** and **360**, respectively, as shown in FIG. **17**.

[0223] In embodiments where the vehicle waste management system **100** includes the vacuum pump system **294** installed with the waste container **182**, the inlet end interlocking portions **330** and **332** of the upstream valve apparatus **298** may be releasably interlocked with corresponding interlocking portions (not shown) on the pump system inlet fitting **216** to releasably secure and connect the inlet end **308** of the upstream valve apparatus **298** to the pump system inlet fitting **216**, the outlet end interlocking portions **334** and **336** of the upstream valve apparatus **298** are releasably interlocked with the vacuum pump inlet interlocking portions **342** and **344** of the vacuum pump **296** to releasably secure and connect the outlet end **310** of the upstream valve apparatus **298** to the pump inlet **302**, the inlet end interlocking portions **346** and **348** of the downstream valve apparatus **300** are releasably interlocked with the vacuum pump outlet interlocking portions **358** and **360** of the vacuum pump **296** to releasably secure and connect the inlet end **320** of the downstream valve apparatus **300** to the pump outlet **304**, and the outlet end interlocking portions **350** and **352** of the downstream valve apparatus **300** may be releasably interlocked with corresponding interlocking portions (not shown) on the pump system outlet fitting **218** to releasably secure and connect the

outlet end **322** of the downstream valve apparatus **300** to the pump system outlet fitting **218**. Thus, the upstream valve apparatus **298** may be connected and releasably secured to the pump system inlet fitting **216** and to the vacuum pump **296**, and the downstream valve apparatus **300** may be connected and releasably secured to the vacuum pump **296** and to the pump system outlet fitting **218**.

[0224] Of course, the embodiment shown is an example only, and alternative embodiments may differ. For example, in the embodiment shown, the upstream and downstream valve apparatuses **298** and **300** each include two inlet end interlocking portions (i.e., **330/346** and **332/348**) and two outlet end interlocking portions (i.e., **334/350** and **336/352**). However, alternative embodiments may include valve apparatuses having only a single inlet end interlocking portion and/or a single outlet end interlocking portion, or valve apparatuses having more than two inlet end interlocking portions and/or more than two outlet end interlocking portions. Additionally, in some alternative embodiments, at least one of the inlet/outlet end interlocking portions may project into the flow channel of the valve apparatus, rather than away from the flow channel.

[0225] FIGS. **18** and **19** show the upstream valve apparatus **298** in greater detail. However, the following explanation and references to FIGS. **18** and **19** are also applicable to the downstream valve apparatus **300**, which, as noted above, is generally similar to the upstream valve apparatus **298**. Referring to FIGS. **18** and **19**, the inlet end **308** of the upstream valve apparatus **298** includes an inlet end mating surface **362** around the channel inlet **312**, and the outlet end **310** of the upstream valve apparatus **298** includes an outlet end mating surface **364** around the channel outlet **314**. Generally, the inlet end mating surface **362** is configured to engage with a corresponding waste source mating surface of a waste source of the upstream valve apparatus **298** when the inlet end **308** is connected to the waste source. Similarly, the outlet end mating surface **364** is configured to engage with a corresponding waste destination mating surface of a waste destination of the upstream valve apparatus **298** when the outlet end **310** is connected to the waste destination. In the embodiment shown, the inlet end mating surface **362** may engage with a corresponding mating surface (not shown) of the pump system inlet fitting **216**, and the outlet end mating surface **364** engages with a corresponding pump inlet mating surface (not shown) of the vacuum pump **296**. However, alternative embodiments may differ. For example, in some alternative embodiments, the inlet end mating surface **362** may engage with a corresponding mating surface that is directly on the top wall **192** of the waste container **182**.

[0226] The upstream valve apparatus **298** also includes channel inlet seals **366** and **368** seated on the inlet end mating surface **362** around the channel inlet **312**, and a channel outlet seals **370** and **372** seated on the outlet end mating surface **364** around the channel outlet **314**. Generally, the channel inlet seals **366** and **368** are configured to seal an interface between the inlet end mating surface **362** and a corresponding waste source mating surface of a waste source of the upstream valve apparatus **298** when the inlet end mating surface **362** engages with the waste source mating surface, and the channel outlet seals **370** and **372** are configured to seal an interface between the outlet end mating surface **364** and a corresponding waste destination mating surface of a waste destination of the upstream valve apparatus **298** when the outlet end mating surface **364** engages with the waste destination mating surface. In the embodiment shown, the channel inlet seals **366** and **368** may seal an interface between the inlet end mating surface **362** and the corresponding mating surface of the pump system inlet fitting **216**, and the channel outlet seals **370** and **372** seal an interface between the outlet end mating surface **364** and the corresponding pump inlet mating surface of the vacuum pump **296**. In the embodiment shown, each of the channel inlet seals **366** and **368** and the channel outlet seals **370** and **372** includes an O-ring. However, the embodiment shown is an example only, and alternative embodiments may differ. For example, some alternative embodiments may include inlet/outlet seals which include a face seal or another type of mechanical seal.

[0227] The upstream valve apparatus **298** also includes a first check valve **374** and a second check

valve **376**. The first check valve **374** and the second check valve **376** are retained and secured within the flow channel **316** by the valve body **306**. Each of the first check valve **374** and the second check valve **376** is configured to allow waste to flow through the flow channel **316** in a forward flow direction **378** from the channel inlet **312** to the channel outlet **314**, and to prevent the waste from flowing through the flow channel **316** in a reverse flow direction **380** from the channel outlet **314** to the channel inlet **312**. Thus, the upstream valve apparatus **298** is generally configured to allow waste to flow from the channel inlet **312** to the channel outlet **314** (in the forward flow direction **378**), and to prevent waste from flowing from the channel outlet **314** to the channel inlet **312** (in the reverse flow direction **380**). Similarly, the downstream valve apparatus **300** is generally configured to allow waste to flow from the channel inlet **324** to the channel outlet **326**, and to prevent waste from flowing from the channel outlet **326** to the channel inlet **324**. Consequently, when the upstream valve apparatus **298** and the downstream valve apparatus **300** are connected and releasably secured to the vacuum pump **296** as shown in FIG. 17, such that the flow channel **316** of the upstream valve apparatus **298** is in fluid communication with the vacuum pump **296** through the pump inlet **302** and the flow channel **328** of the downstream valve apparatus **300** is in fluid communication with the vacuum pump **296** through the pump outlet **304**, the upstream valve apparatus **298** generally allows waste to flow into the vacuum pump **296** through the pump inlet **302** while preventing waste from flowing out of the vacuum pump **296** through the pump inlet **302**, and the downstream valve apparatus **300** generally allows waste to flow out of the vacuum pump **296** through the pump outlet **304** while preventing waste from flowing into the vacuum pump **296** through the pump outlet **304**. Therefore, the upstream valve apparatus **298** and the downstream valve apparatus **300** effectively control the direction of flow of waste through the vacuum pump **296** when the vacuum pump **296** causes the waste to flow through the vacuum pump system **294**, generally allowing the waste to flow in a direction from the pump inlet **302** to the pump outlet **304**, and generally preventing the waste from flowing in a direction from the pump outlet **304** to the pump inlet **302**. Thus, overall, when the vacuum pump **296** causes waste to flow through the vacuum pump system **294**, the waste may flow through the vacuum pump system **294** by entering the upstream valve apparatus **298** (and thus the vacuum pump system **294**) through the channel inlet **312**, passing through the flow channel **316**, leaving the upstream valve apparatus **298** through the channel outlet **314**, entering the vacuum pump **296** through the pump inlet **302**, passing through the vacuum pump **296**, leaving the vacuum pump **296** through the pump outlet **304**, entering the downstream valve apparatus **300** through the channel inlet **324**, passing through the flow channel **328**, and leaving the downstream valve apparatus **300** (and thus the vacuum pump system **294**) through the channel outlet **326**.

[0228] In the embodiment shown, each of the first check valve **374** and the second check valve **376** of the upstream valve apparatus **298** is a duckbill valve. Also, as shown in FIG. 19, the first check valve **374** and the second check valve **376** are non-overlapping in the forward flow direction **378** (and, similarly, are non-overlapping in the reverse flow direction **380**). Thus, unlike the first check valve **278** and the second check valve **280** of the embodiment of FIGS. 9 to 14, the first check valve **374** and the second check valve **376** are not nested. However, the embodiment shown is an example only, and alternative embodiments may differ. For example, some alternative embodiments may include other types of check valves instead of or in addition to duckbill valves. Additionally, in some alternative embodiments, the valve apparatus may include a different number of check valves in the flow channel, such as only a single check valve in the flow channel, or three or more check valves in the flow channel.

[0229] In the embodiment shown, the valve body **306** of the upstream valve apparatus **298** includes valve body portions **382**, **384**, and **386** arranged between the channel inlet **312** and the channel outlet **314**. The valve body portions **382**, **384**, and **386** collectively define the flow channel **316** and retain the first and second check valves **374** and **376** in the flow channel **316**. The valve body portion **382** is non-dismantably fixed to the valve body portion **384**, which is in turn non-

dismantlably fixed to the valve body portion **386**. That is, each of the valve body portions **382**, **384**, and **386** is non-dismantlably fixed to at least one other of the valve body portions **382**, **384**, and **386**, such that the valve body **306** as a whole is non-dismantlable. The valve body portions **382**, **384**, and **386** may be fixed together by, for example, plastic welding and/or glue. Of course, the embodiment shown is an example only, and alternative embodiments may differ. For example, in alternative embodiments, the valve body **306** may include a different number of valve body portions non-dismantlably fixed together.

[0230] Generally, the upstream valve apparatus **298**, or a similar valve apparatus such as the downstream valve apparatus **300**, may be installed on to a waste source (such as the pump system inlet fitting **216** of the waste container **182**) by positioning the inlet end **308** at the waste source and rotating the upstream valve apparatus **298**, relative to the to the waste source, around the inlet channel axis **338** to cause the inlet end interlocking portions **330** and **332** to slide against and releasably interlock with corresponding waste source interlocking portions of the waste source. The upstream valve apparatus **298** may then be uninstalled from the waste source by further rotating the upstream valve apparatus **298**, relative to the to the waste source, around the inlet channel axis **338** to release the interlock between the inlet end interlocking portions **330** and **332** and the corresponding waste source interlocking portions. Similarly, the upstream valve apparatus **298** may be installed on to a waste destination (such as the pump inlet **302** of the vacuum pump **296**) by positioning the outlet end **310** at the waste destination and rotating the upstream valve apparatus **298**, relative to the to the waste destination, around the outlet channel axis **340** to cause the outlet end interlocking portions **334** and **336** to slide against and releasably interlock with corresponding waste destination interlocking portions of the waste destination. The upstream valve apparatus **298** may then be uninstalled from the waste destination by further rotating the upstream valve apparatus **298**, relative to the to the waste destination, around the outlet channel axis **340** to release the interlock between the outlet end interlocking portions **334** and **336** and the corresponding waste destination interlocking portions. That is, installation and uninstallation of the upstream valve apparatus **298**, or of a similar valve apparatus such as the downstream valve apparatus **300**, require only the positioning and rotation of the upstream valve apparatus **298** via the twist lock securing mechanism. As such, valve apparatuses such as the upstream valve apparatus **298** and the downstream valve apparatus **300** disclosed herein may simplify maintenance (e.g., valve replacement) of waste management systems such as the waste management system **100**.

[0231] Referring back to FIGS. **9**, **10**, **11**, **15**, **16**, and **17**, when installed on the waste container **182** (e.g., as shown in FIGS. **9** and **10**), the vacuum pump system **183** or **294** is generally operable to cause waste and other fluids, including, for example, air, to flow from the vacuum chamber **202** to the main chamber **198**. Thus, if the top vacuum chamber inlet **204** is sealed (e.g., by a valve) or is in fluid communication through the inlet pipe **214** with a sealed environment (e.g., piping from a toilet and/or sink), the vacuum pump system **183** or **294** is operable to purge or evacuate the vacuum chamber **202** to build up a vacuum charge in the vacuum chamber **202**. When the waste container **182** and the vacuum pump system **183** or **294** are installed on the watercraft **102** and the inlet pipe **214** is in fluid communication with a waste source such as a toilet and/or sink onboard the watercraft **102**, such a vacuum charge in the vacuum chamber **202** may be used to pull waste and/or other fluids from the toilet or sink into the vacuum chamber **202**. For example, if the inlet pipe **214** is in fluid communication with piping from a toilet, and the piping is releasably sealed between the toilet and the inlet pipe **214** to allow a vacuum charge to be built up in the vacuum chamber **202**, when a user flushes the toilet, the seal may be temporarily disengaged such that the vacuum charge may pull any waste from the toilet into the vacuum chamber **202**. The vacuum pump system **183** or **294** may then transfer this additional waste from the vacuum chamber **202** to the main chamber **198**, and (once the releasable seal in the piping has been re-engaged) may once again build up the vacuum charge in the vacuum chamber **202** in preparation for a subsequent use.

[0232] The following non-limiting examples are illustrative of embodiments of the present

disclosure.

Example 1: Dye Dispersion Test

[0233] In this example, a dye was used to assess dispersion of fluids during tilting motion in waste containers similar to the waste container **104** of the embodiment of FIGS. **2** to **8**, having different configurations of projections similar to the projections **128, 130, 132, 134, 136, 138, 140, 142, 144, 146, 148, 150, 152, 154, 156, 158, 160, 162, 164, 166, 168, 170, 172, and 174**. More specifically, dye dispersion during tilting motion was assessed in waste containers having: [0234] a) no projections (control); [0235] b) parallel projections spaced apart from each other at a fixed projection spacing of 2 inches and having a depth of 1.25 inches; [0236] c) parallel projections spaced apart from each other at a fixed projection spacing of 1 inch and having a depth of 1.25 inches; [0237] d) parallel projections spaced apart from each other at a fixed projection spacing of 1 inch and having alternating depths of 0.5 and 1.25 inches; [0238] e) parallel projections spaced apart from each other at a fixed projection spacing of 2 inches and having a depth of 0.5 inches; and [0239] f) parallel projections spaced apart from each other at a fixed projection spacing of 1 inch and having a depth of 0.5 inches.

[0240] In each case, a dye was added to the waste container, tilting motion of the waste container was initiated, and a time to complete dispersion of the dye was measured. The results of these measurements are summarized in Table 1.

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TABLE 1	Time to Complete Dispersion of Dye During Tilting Motion	Waste Container Configuration	Time to Complete Dispersion (sec)
No projections (control)	91	1.25" projections, 2" spacing	15.25
1.25" projections, 1" spacing	12.5	1.25"/0.5" projections, 1" spacing	15
0.5" projections, 2" spacing	30.5	0.5" projections, 1" spacing	32.75

[0241] As shown in Table 1, without projections (control configuration), complete dispersion requires approximately a minute and a half, while with 1.25 inch projections, complete dispersion requires only 12 to 15 seconds. This represents an approximate 80% decrease in time to complete dispersion.

[0242] The effect of decreasing projection spacing, and thus increasing projection density, was less pronounced. A drop from 2 inch spacing to 1 inch spacing, representing a doubling of the total number of projections, provided only a limited increase in effectiveness for a large increase in material use, tooling cost, and complexity of production. However, the effect of increasing the projection depth from 0.5 inches to 1.25 inches yielded a much larger increase in turbulent flow, dispersion, and mixing of the fluid in the tank.

Example 2: Solids Breakdown Test

[0243] In this example, mixtures of water and macerated marine toilet paper were used to assess and compare capacities for dispersing suspended solids during motion in a waste container having no projections (control) and in a waste container with parallel projections spaced apart from each other at a fixed projection spacing of 2 inches and having a depth of 1.25 inches. In each case, the mixture was added to the respective container and allowed to reach equilibrium, with the macerated toilet paper settling at the bottom of the container. Then, the container was subjected to a tilting/sloshing motion, and the behaviour of the mixture inside was recorded. In the control container without projections, little to no movement of the macerated toilet paper was observed, and the macerated toilet paper remained at the bottom of the container for an entire duration of the test (approximately 112minutes). Conversely, in the container with 1.25 inch projections spaced at 2 inches, within a minute of sloshing, the macerated toilet paper was observed to have fully dispersed in the container and become suspended in the water. The mixture remained this way for the remainder of the test. It was also observed that turbulent flow created between each projection appeared to create a mixing action, which appeared to resuspend previously suspended solids at the bottom of the container when the tilting motion was applied.

[0244] These results appear to show that, compared to a waste container having no projections, a waste container having projections such as the projections **128, 130, 132, 134, 136, 138, 140, 142,**

144, 146, 148, 150, 152, 154, 156, 158, 160, 162, 164, 166, 168, 170, 172, and 174 of the waste container 104 of the embodiment of FIGS. 2 to 8 increases a dispersion of suspended solids in the container and prevents settling of semi-suspended solids.

Clauses

[0245] This disclosure includes but is not limited to the following clauses, which may be combined with other subject matter in this specification. [0246] 1. A waste container for use on a vehicle, the waste container comprising:

[0247] at least one wall comprising an inner surface defining a chamber for holding waste; and

[0248] a plurality of projections projecting from the inner surface of the at least one wall into the chamber. [0249] 2. The waste container of clause 1 wherein the projections are configured to

disturb flow of the waste in the chamber as movement of the vehicle causes the waste in the chamber to move in the chamber. [0250] 3. The waste container of clause 2 wherein the projections

are configured to generate turbulence in the waste in the chamber as the movement of the vehicle causes the waste in the chamber to move in the chamber. [0251] 4. The waste container of clause 2

or 3 wherein the projections are configured to break apart solids in the waste in the chamber as the movement of the vehicle causes the waste in the chamber to move in the chamber. [0252] 5. The

waste container of clause 2, 3, or 4 wherein the projections are configured to disperse particles in the waste in the chamber as the movement of the vehicle causes the waste in the chamber to move

in the chamber. [0253] 6. The waste container of any one of clauses 2 to 5 wherein the movement of the vehicle comprises tilting of the vehicle. [0254] 7. The waste container of any one of clauses

1 to 6 wherein at least one of the plurality of projections extends along the inner surface without contacting any other one of the plurality of projections. [0255] 8. The waste container of any one of

clauses 1 to 7 wherein each of the plurality of projections extends along the inner surface without contacting any other one of the plurality of projections. [0256] 9. The waste container of any one of

clauses 1 to 8 wherein at least some of the plurality of projections extend along the inner surface parallel to one another. [0257] 10. The waste container of clause 9 wherein adjacent ones of the

plurality of projections along the inner surface, which extend parallel to one another along the inner surface, are spaced apart from one another at a fixed projection spacing in a spacing direction along

the inner surface, the spacing direction perpendicular to an extension direction along the inner surface of the adjacent ones of the plurality of projections. [0258] 11. The waste container of clause

10 wherein the projection spacing is at most 1 inch. [0259] 12. The waste container of clause 10 wherein the projection spacing is about 1 inch. [0260] 13. The waste container of clause 10 wherein

the projection spacing is at least 1 inch. [0261] 14. The waste container of clause 10 or 13 wherein the projection spacing is at most 2 inches. [0262] 15. The waste container of clause 10 wherein the

projection spacing is about 2 inches. [0263] 16. The waste container of any one of clauses 1 to 15 wherein each of the plurality of projections extends along the inner surface in a substantially

vertical or vertical direction when the waste container is installed on the vehicle. [0264] 17. The waste container of any one of clauses 1 to 16 wherein each of the plurality of projections projects

from a respective portion of the inner surface in a respective projection direction generally orthogonal or orthogonal to the respective portion of the inner surface. [0265] 18. The waste

container of any one of clauses 1 to 17 wherein at least a portion of each of the plurality of projections projects at least 0.5 inches from the inner surface. [0266] 19. The waste container of

clause 18 wherein all of each of the plurality of projections projects at least 0.5 inches from the inner surface. [0267] 20. The waste container of any one of clauses 1 to 19 wherein at least a

portion of each of the plurality of projections projects about 0.5 inches from the inner surface. [0268] 21. The waste container of clause 20 wherein all of each of the plurality of projections

projects about 0.5 inches from the inner surface. [0269] 22. The waste container of any one of clauses 1 to 19 wherein at least a portion of each of the plurality of projections projects at least 1.25

inches from the inner surface. [0270] 23. The waste container of clause 22 wherein all of each of the plurality of projections projects at least 1.25 inches from the inner surface. [0271] 24. The

waste container of any one of clauses 1 to 19, 22, or 23 wherein at least a portion of each of the plurality of projections projects about 1.25 inches from the inner surface. [0272] 25. The waste container of clause 24 wherein all of each of the plurality of projections projects about 1.25 inches from the inner surface. [0273] 26. The waste container of any one of clauses 1 to 21 wherein each of the plurality of projections projects at most 0.5 inches from the inner surface. [0274] 27. The waste container of any one of clauses 1 to 25 wherein each of the plurality of projections projects at most 1.25 inches from the inner surface. [0275] 28. The waste container of any one of clauses 1 to 27 wherein the at least one wall further defines at least one opening in the chamber for conveying the waste into and out of the chamber. [0276] 29. The waste container of clause 28 wherein the at least one opening is closable. [0277] 30. The waste container of clause 28 or 29 wherein at least one of the at least one opening is at a top of the chamber or a bottom of the chamber when the waste container is installed on the vehicle. [0278] 31. The waste container of clause 28, 29, or 30 wherein the at least one opening comprises:

[0279] at least one inlet for receiving the waste into the chamber; and

[0280] at least one outlet for removing the waste from the chamber. [0281] 32. The waste container of clause 31, when dependent from clause 30, wherein the at least one outlet comprises a bottom outlet positioned to be at the bottom of the chamber when the waste container is installed on the vehicle. [0282] 33. The waste container of clause 32 further comprising a valve operable to reversibly close the bottom outlet. [0283] 34. The waste container of clause 33 wherein the valve is a ball valve. [0284] 35. The waste container of clause 32, 33, or 34 wherein the bottom outlet is configured to be in fluid communication with a seacock on a hull of the vehicle when the waste container is installed on the vehicle. [0285] 36. The waste container of clause 31, when dependent from clause 30, or of any one of clauses 32 to 35 wherein the at least one outlet comprises a top outlet positioned to be at the top of the chamber when the waste container is installed on the vehicle. [0286] 37. The waste container of clause 36 further comprising a dip tube extending from the top outlet into the chamber, the dip tube in fluid communication with the top outlet and with the chamber. [0287] 38. The waste container of clause 36 or 37 wherein the top outlet is configured to be in fluid communication with a pump-out fitting of the vehicle when the waste container is installed on the vehicle. [0288] 39. The waste container of clause 31, when dependent from clause 30, or of any one of clauses 32 to 38 wherein the at least one inlet comprises a top inlet positioned to be at the top of the chamber when the waste container is installed on the vehicle. [0289] 40. The waste container of any one of clauses 31 to 39 wherein the at least one inlet is configured to be in fluid communication with a toilet onboard the vehicle when the waste container is installed on the vehicle. [0290] 41. The waste container of any one of clauses 31 to 40 wherein the at least one inlet is configured to be in fluid communication with a sink onboard the vehicle when the waste container is installed on the vehicle. [0291] 42. The waste container of any one of clauses 31 to 41 wherein the at least one inlet is configured to be in fluid communication with a shower onboard the vehicle when the waste container is installed on the vehicle. [0292] 43. The waste container of any one of clauses 31 to 42 wherein each of the plurality of projections extends along at least a portion of the inner surface between one of the at least one inlet and one of the at least one outlet. [0293] 44. The waste container of any one of clauses 1 to 43 wherein the at least one wall comprises a plurality of walls. [0294] 45. The waste container of clause 44 wherein at least some of the plurality of walls are parallel to one another. [0295] 46. The waste container of clause 44 or 45 wherein at least one of the plurality of projections projects from each of the plurality of walls. [0296] 47. The waste container of clause 44 or 45 wherein none of the plurality of projections project from at least one of the plurality of walls. [0297] 48. The waste container of any one of clauses 1 to 47 wherein the vehicle is a marine vessel. [0298] 49. The waste container of any one of clauses 1 to 48 installed on the vehicle. [0299] 50. A method of manufacturing the waste container of any one of clauses 1 to 48, the method comprising molding the at least one wall with the plurality of projections. [0300] 51. A method of manufacturing the waste container of any one of

clauses 1 to 48, the method comprising:
[0301] molding the at least one wall; and
[0302] fixing the plurality of projections to the inner surface. [0303] 52. The method of clause 50 or 51 wherein molding the at least one wall comprises rotational molding the at least one wall.
[0304] 53. The method of clause 50, 51, or 52 wherein molding the at least one wall comprises injection molding the at least one wall. [0305] 54. A system comprising:
[0306] the waste container of any one of clauses 1 to 48; and
[0307] the vehicle,
[0308] wherein the waste container is installed on the vehicle. [0309] 55. Use of the waste container of any one of clauses 1 to 48 for waste management on the vehicle. [0310] 56. A valve apparatus for controlling flow of waste into or out of a waste container, the valve apparatus comprising:
[0311] a non-dismantlable valve body defining a flow channel for conveying the waste, the flow channel comprising a channel inlet for receiving the waste into the flow channel and a channel outlet for removing the waste from the flow channel; and
[0312] at least one check valve secured within the flow channel, the at least one check valve configured to: [0313] allow the waste to flow through the flow channel in a forward flow direction from the channel inlet to the channel outlet; and [0314] prevent the waste from flowing through the flow channel in a reverse flow direction from the channel outlet to the channel inlet. [0315] 57. The valve apparatus of clause 56 for use with a pump for controlling the flow of the waste into or out of the waste container. [0316] 58. The valve apparatus of clause 57 wherein the pump comprises a vacuum pump. [0317] 59. The valve apparatus of clause 56, 57, or 58 wherein the valve body comprises an inlet end defining the channel inlet, the inlet end connectable to a waste source to cause the flow channel to be in fluid communication with the waste source through the channel inlet. [0318] 60. The valve apparatus of clause 59 further comprising a channel inlet clamp operable to releasably secure the inlet end to the waste source to connect the inlet end to the waste source to cause the flow channel to be in fluid communication with the waste source through the channel inlet. [0319] 61. The valve apparatus of clause 59 or 60 wherein the inlet end comprises at least one inlet end interlocking portion, the at least one inlet end interlocking portion releasably interlockable with a corresponding waste source interlocking portion of the waste source to releasably secure the inlet end to the waste source to connect the inlet end to the waste source to cause the flow channel to be in fluid communication with the waste source through the channel inlet. [0320] 62. The valve apparatus of clause 61 wherein the at least one inlet end interlocking portion is releasably interlockable with the waste source interlocking portion by rotation of the valve apparatus relative to the waste source. [0321] 63. The valve apparatus of clause 62 wherein the rotation of the valve apparatus relative to the waste source is around an inlet channel axis parallel to the flow channel at the channel inlet. [0322] 64. The valve apparatus of clause 61, 62, or 63 wherein the at least one inlet end interlocking portion projects from the valve body at the inlet end. [0323] 65. The valve apparatus of clause 64 wherein the at least one inlet end interlocking portion projects out from the valve body away from the flow channel. [0324] 66. The valve apparatus of any one of clauses 59 to 65 wherein the inlet end comprises an inlet end mating surface around the channel inlet, the inlet end mating surface configured to engage with a corresponding waste source mating surface of the waste source when the inlet end is connected to the waste source to cause the flow channel to be in fluid communication with the waste source through the channel inlet. [0325] 67. The valve apparatus of clause 66 further comprising a channel inlet seal seated on the inlet end mating surface around the channel inlet, the channel inlet seal configured to seal an interface between the inlet end mating surface and the waste source mating surface when the inlet end mating surface engages with the waste source mating surface. [0326] 68. The valve apparatus of clause 67 wherein the channel inlet seal comprises a face seal. [0327] 69. The valve apparatus of clause 67 or 68 wherein the channel inlet seal comprises an O-ring. [0328]

70. The valve apparatus of any one of clauses 59 to 69 wherein the waste source comprises the waste container. [0329] 71. The valve apparatus of any one of clauses 59 to 69 when directly or indirectly dependent from clause 57 wherein the waste source comprises the vacuum pump. [0330] 72. The valve apparatus of any one of clauses 56 to 71 wherein the valve body comprises an outlet end defining the channel outlet, the outlet end connectable to a waste destination to cause the flow channel to be in fluid communication with the waste destination through the channel outlet. [0331] 73. The valve apparatus of clause 72 further comprising a channel outlet clamp operable to releasably secure the outlet end to the waste destination to connect the outlet end to the waste destination to cause the flow channel to be in fluid communication with the waste destination through the channel outlet. [0332] 74. The valve apparatus of clause 72 or 73 wherein the outlet end comprises at least one outlet end interlocking portion, the at least one outlet end interlocking portion releasably interlockable with a corresponding waste destination interlocking portion of the waste destination to releasably secure the outlet end to the waste destination to connect the outlet end to the waste destination to cause the flow channel to be in fluid communication with the waste destination through the channel outlet. [0333] 75. The valve apparatus of clause 74 wherein the at least one outlet end interlocking portion is releasably interlockable with the waste destination interlocking portion by rotation of the valve apparatus relative to the waste destination. [0334] 76. The valve apparatus of clause 75 wherein the rotation of the valve apparatus relative to the waste destination is around an outlet channel axis parallel to the flow channel at the channel outlet. [0335] 77. The valve apparatus of clause 74, 75, or 76 wherein the at least one outlet end interlocking portion projects from the valve body at the outlet end. [0336] 78. The valve apparatus of clause 77 wherein the at least one outlet end interlocking portion projects out from the valve body away from the flow channel. [0337] 79. The valve apparatus of any one of clauses 72 to 78 wherein the outlet end comprises an outlet end mating surface around the channel outlet, the outlet end mating surface configured to engage with a corresponding waste destination mating surface of the waste destination when the outlet end is connected to the waste destination to cause the flow channel to be in fluid communication with the waste destination through the channel outlet. [0338] 80. The valve apparatus of clause 79 further comprising a channel outlet seal seated on the outlet end mating surface around the channel outlet, the channel outlet seal configured to seal an interface between the outlet end mating surface and the waste destination mating surface when the outlet end mating surface engages with the waste destination mating surface. [0339] 81. The valve apparatus of clause 80 wherein the channel outlet seal comprises a face seal. [0340] 82. The valve apparatus of clause 80 or 81 wherein the channel outlet seal comprises an O-ring. [0341] 83. The valve apparatus of any one of clauses 72 to 82 wherein the waste destination comprises the waste container. [0342] 84. The valve apparatus of any one of clauses 72 to 82 when directly or indirectly dependent from clause 57 wherein the waste destination comprises the vacuum pump. [0343] 85. The valve apparatus of any one of clauses 56 to 84 wherein the valve body comprises a plurality of valve body portions between the channel inlet and the channel outlet, the plurality of valve body portions collectively defining the flow channel, each valve body portion of the plurality of valve body portions fixed to at least one other valve body portion of the plurality of valve body portions. [0344] 86. The valve apparatus of clause 85 wherein at least two of the valve body portions of the plurality of valve body portions are fixed together by plastic welding. [0345] 87. The valve apparatus of clause 85 or 86 wherein at least two of the valve body portions of the plurality of valve body portions are fixed together by glue. [0346] 88. The valve apparatus of clause 85, 86, or 87 wherein at least one valve body portion of the plurality of valve body portions retains the check valve. [0347] 89. The valve apparatus of any one of clauses 56 to 88 wherein the at least one check valve comprises at least one duckbill valve. [0348] 90. The valve apparatus of clause 89 wherein the at least one duckbill valve comprises a first duckbill valve and a second duckbill valve. [0349] 91. The valve apparatus of clause 90 wherein the first duckbill valve and the second duckbill valve are non-overlapping in the forward flow direction. [0350] 92. The valve apparatus of clause 90

wherein at least a portion of the first duckbill valve is received in the second duckbill valve. [0351] 93. The valve apparatus of clause 92 wherein the first duckbill valve is nested within the second duckbill valve. [0352] 94. The valve apparatus of any one of clauses 56 to 93 for controlling the flow of the waste into or out of the waste container of any one of clauses 1 to 48. [0353] 95. A method of installing the valve apparatus of clause 60 or of any one of clauses 61 to 93 when directly or indirectly dependent from clause 60 to the waste source, the method comprising: [0354] causing the channel inlet clamp to secure the inlet end to the waste source. [0355] 96. A method of installing the valve apparatus of clause 62 or 63 or of any one of clauses 64 to 93 when directly or indirectly dependent from clause 62 to the waste source, the method comprising: [0356] rotating the valve apparatus relative to the waste source to interlock the at least one inlet end interlocking portion with the waste source interlocking portion. [0357] 97. A method of installing the valve apparatus of clause 73 or of any one of clauses 74 to 93 when directly or indirectly dependent from clause 73 to the waste destination, the method comprising: [0358] causing the channel outlet clamp to secure the outlet end to the waste destination. [0359] 98. A method of installing the valve apparatus of clause 75 or 76 or of any one of clauses 77 to 93 when directly or indirectly dependent from clause 75 to the waste destination, the method comprising: [0360] rotating the valve apparatus relative to the waste destination to interlock the at least one outlet end interlocking portion with the waste destination interlocking portion. [0361] 99. A system comprising: [0362] the valve apparatus of any one of clauses 56 to 93; and [0363] the waste container, [0364] wherein: [0365] the waste container comprises at least one wall comprising an inner surface defining a chamber for holding the waste; and [0366] the flow channel of the valve apparatus is in fluid communication with the chamber of the waste container. [0367] 100. The system of clause 99 wherein: [0368] the at least one wall of the waste container defines a container outlet for removing the waste from the waste container; and [0369] the flow channel of the valve apparatus is in fluid communication with the chamber of the waste container through the container outlet of the waste container and through the channel inlet of the valve apparatus. [0370] 101. The system of clause 100 when directly or indirectly dependent from clause 59 wherein the inlet end of the valve apparatus is connected to the waste container at the container outlet of the waste container. [0371] 102. The system of clause 101 when directly or indirectly dependent from clause 60 wherein the channel inlet clamp of the valve apparatus releasably secures the inlet end of the valve apparatus to the waste container at the container outlet of the waste container. [0372] 103. The system of clause 101 or 102 when directly or indirectly dependent from clause 61 wherein the waste container further comprises at least one waste container outlet interlocking portion at the container outlet, the at least one waste container outlet interlocking portion releasably interlocked with a corresponding one of the at least one inlet end interlocking portion of the inlet end of the valve apparatus to releasably secure the inlet end of the valve apparatus to the waste container at the container outlet of the waste container. [0373] 104. The system of clause 101, 102, or 103 when directly or indirectly dependent from clause 66 wherein the inlet end mating surface of the valve apparatus engages with a corresponding container outlet mating surface of the waste container. [0374] 105. The system of clause 104 when directly or indirectly dependent from clause 67 wherein the channel inlet seal of the valve apparatus seals an interface between the inlet end mating surface of the valve apparatus and the container outlet mating surface of the waste container. [0375] 106. The system of clause 99 wherein: [0376] the at least one wall of the waste container defines a container inlet for receiving the waste into the waste container; and [0377] the flow channel of the valve apparatus is in fluid communication with the waste container

through the container inlet of the waste container and through the channel outlet of the valve apparatus. [0378] 107. The system of clause 106 when directly or indirectly dependent from clause 72 wherein the outlet end of the valve apparatus is connected to the waste container at the container inlet of the waste container. [0379] 108. The system of clause 107 when directly or indirectly dependent from clause 73 wherein the channel outlet clamp of the valve apparatus releasably secures the outlet end of the valve apparatus to the waste container at the container inlet of the waste container. [0380] 109. The system of clause 107 or 108 when directly or indirectly dependent from clause 74 wherein the waste container further comprises at least one waste container inlet interlocking portion at the container inlet, the at least one waste container inlet interlocking portion releasably interlocked with a corresponding one of the at least one outlet end interlocking portion of the outlet end of the valve apparatus to releasably secure the outlet end of the valve apparatus to the waste container at the container inlet of the waste container. [0381] 110. The system of clause 107, 108, or 109 when directly or indirectly dependent from clause 79 wherein the outlet end mating surface of the valve apparatus engages with a corresponding container inlet mating surface of the waste container. [0382] 111. The system of clause 110 when directly or indirectly dependent from clause 80 wherein the channel outlet seal of the valve apparatus seals an interface between the outlet end mating surface of the valve apparatus and the container inlet mating surface of the waste container. [0383] 112. The system of any one of clauses 99 to 111 wherein the waste container comprises the waste container of any one of clauses 1 to 48. [0384] 113. A vacuum pump system for controlling flow of waste into or out of a waste container, the vacuum pump system comprising: [0385] a vacuum pump operable to cause the waste to flow, the vacuum pump defining a pump inlet for receiving the waste into the vacuum pump and a pump outlet for removing the waste from the vacuum pump; and [0386] a first one of the valve apparatus of any one of clauses 56 to 93, [0387] wherein the flow channel of the first one of the valve apparatus is in fluid communication with the vacuum pump. [0388] 114. The vacuum pump system of clause 113 wherein the flow channel of the first one of the valve apparatus is in fluid communication with the vacuum pump through the pump outlet of the vacuum pump and through the channel inlet of the first one of the valve apparatus. [0389] 115. The vacuum pump system of clause 114 when directly or indirectly dependent from clause 59 wherein the inlet end of the first one of the valve apparatus is connected to the vacuum pump at the pump outlet of the vacuum pump. [0390] 116. The vacuum pump system of clause 115 when directly or indirectly dependent from clause 60 wherein the channel inlet clamp of the first one of the valve apparatus releasably secures the inlet end of the first one of the valve apparatus to the vacuum pump at the pump outlet of the vacuum pump. [0391] 117. The vacuum pump system of clause 115 or 116 when directly or indirectly dependent from clause 61 wherein the vacuum pump further comprises at least one vacuum pump outlet interlocking portion at the pump outlet, the at least one vacuum pump outlet interlocking portion releasably interlocked with a corresponding one of the at least one inlet end interlocking portion of the inlet end of the first one of the valve apparatus to releasably secure the inlet end of the first one of the valve apparatus to the vacuum pump at the pump outlet of the vacuum pump. [0392] 118. The vacuum pump system of clause 115, 116, or 117 when directly or indirectly dependent from clause 66 wherein the inlet end mating surface of the first one of the valve apparatus engages with a corresponding pump outlet mating surface of the vacuum pump. [0393] 119. The vacuum pump system of clause 118 when directly or indirectly dependent from clause 67 wherein the channel inlet seal of the first one of the valve apparatus seals an interface between the inlet end mating surface of the first one of the valve apparatus and the pump outlet mating surface of the vacuum pump. [0394] 120. The vacuum pump system of clause 113 wherein the flow channel of the first one of the valve apparatus is in fluid communication with the vacuum pump through the pump inlet of the vacuum pump and through the channel outlet of the first one of the valve apparatus. [0395] 121. The vacuum pump system of clause 120 when directly or indirectly dependent from clause 72

wherein the outlet end of the first one of the valve apparatus is connected to the vacuum pump at the pump inlet of the vacuum pump. [0396] 122. The vacuum pump system of clause 121 when directly or indirectly dependent from clause 73 wherein the channel outlet clamp of the first one of the valve apparatus releasably secures the outlet end of the first one of the valve apparatus to the vacuum pump at the pump inlet of the vacuum pump. [0397] 123. The vacuum pump system of clause 121 or 122 when directly or indirectly dependent from clause 74 wherein the vacuum pump further comprises at least one vacuum pump inlet interlocking portion at the pump inlet, the at least one vacuum pump inlet interlocking portion releasably interlocked with a corresponding one of the at least one outlet end interlocking portion of the outlet end of the first one of the valve apparatus to releasably secure the outlet end of the first one of the valve apparatus to the vacuum pump at the pump inlet of the vacuum pump. [0398] 124. The vacuum pump system of clause 121, 122, or 123 when directly or indirectly dependent from clause 79 wherein the outlet end mating surface of the first one of the valve apparatus engages with a corresponding pump inlet mating surface of the vacuum pump. [0399] 125. The vacuum pump system of clause 124 when directly or indirectly dependent from clause 80 wherein the channel outlet seal of the first one of the valve apparatus seals an interface between the outlet end mating surface of the first one of the valve apparatus and the pump inlet mating surface of the vacuum pump. [0400] 126. The vacuum pump system of any one of clauses 120 to 125 further comprising a second one of the valve apparatus of any one of clauses 56 to 93, wherein the flow channel of the second one of the valve apparatus is in fluid communication with the vacuum pump through the pump outlet of the vacuum pump and through the channel inlet of the second one of the valve apparatus. [0401] 127. The vacuum pump system of clause 126 when directly or indirectly dependent from clause 59 wherein the inlet end of the second one of the valve apparatus is connected to the vacuum pump at the pump outlet of the vacuum pump. [0402] 128. The vacuum pump system of clause 127 when directly or indirectly dependent from clause 60 wherein the channel inlet clamp of the second one of the valve apparatus releasably secures the inlet end of the second one of the valve apparatus to the vacuum pump at the pump outlet of the vacuum pump. [0403] 129. The vacuum pump system of clause 128 when directly or indirectly dependent from clause 61 wherein the vacuum pump further comprises at least one vacuum pump outlet interlocking portion at the pump outlet, the at least one vacuum pump outlet interlocking portion releasably interlocked with a corresponding one of the at least one inlet end interlocking portion of the inlet end of the second one of the valve apparatus to releasably secure the inlet end of the second one of the valve apparatus to the vacuum pump at the pump outlet of the vacuum pump. [0404] 130. The vacuum pump system of clause 127, 128, or 129 when directly or indirectly dependent from clause 66 wherein the inlet end mating surface of the second one of the valve apparatus engages with a corresponding pump outlet mating surface of the vacuum pump. [0405] 131. The vacuum pump system of clause 130 when directly or indirectly dependent from clause 67 wherein the channel inlet seal of the second one of the valve apparatus seals an interface between the inlet end mating surface of the second one of the valve apparatus and the pump outlet mating surface of the vacuum pump. [0406] 132. The vacuum pump system of any one of clauses 113 to 131 for controlling the flow of the waste into or out of the waste container of any one of clauses 1 to 48. [0407] 133. A system comprising:
[0408] the vacuum pump system of any one of clauses 113 to 131; and
[0409] the waste container,
[0410] wherein: [0411] the waste container comprises at least one wall comprising an inner surface defining a chamber for holding the waste; and [0412] the vacuum pump system is in fluid communication with the chamber of the waste container. [0413] 134. The system of clause 133 wherein the flow channel of the first one of the valve apparatus of the vacuum pump system is in fluid communication with the chamber of the waste container. [0414] 135. The system of clause 134 when directly or indirectly dependent from clause 114 wherein:
[0415] the at least one wall of the waste container defines a container inlet for receiving the waste

into the waste container; and
[0416] the flow channel of the first one of the valve apparatus of the vacuum pump system is in fluid communication with the chamber of the waste container through the container inlet of the waste container and through the channel outlet of the first one of the valve apparatus of the vacuum pump system. [0417] 136. The system of clause 135 when directly or indirectly dependent from clause 72 wherein the outlet end of the first one of the valve apparatus of the vacuum pump system is connected to the waste container at the container inlet of the waste container. [0418] 137. The system of clause 136 when directly or indirectly dependent from clause 73 wherein the channel outlet clamp of the first one of the valve apparatus of the vacuum pump system releasably secures the outlet end of the first one of the valve apparatus of the vacuum pump system to the waste container at the container inlet of the waste container. [0419] 138. The system of clause 136 or 137 when directly or indirectly dependent from clause 74 wherein the waste container further comprises at least one waste container inlet interlocking portion at the container inlet, the at least one waste container inlet interlocking portion releasably interlocked with a corresponding one of the at least one outlet end interlocking portion of the outlet end of the first one of the valve apparatus to releasably secure the outlet end of the first one of the valve apparatus to the waste container at the container inlet of the waste container. [0420] 139. The system of clause 136, 137, or 138 when directly or indirectly dependent from clause 79 wherein the outlet end mating surface of the first one of the valve apparatus of the vacuum pump system engages with a corresponding container inlet mating surface of the waste container. [0421] 140. The system of clause 139 when directly or indirectly dependent from clause 80 wherein the channel outlet seal of the first one of the valve apparatus of the vacuum pump system seals an interface between the outlet end mating surface of the first one of the valve apparatus of the vacuum pump system and the container inlet mating surface of the waste container. [0422] 141. The system of clause 134 when directly or indirectly dependent from clause 120 wherein:

[0423] the at least one wall of the waste container defines a container outlet for receiving the waste into the waste container; and

[0424] the flow channel of the first one of the valve apparatus of the vacuum pump system is in fluid communication with the chamber of the waste container through the container outlet of the waste container and through the channel inlet of the first one of the valve apparatus of the vacuum pump system. [0425] 142. The system of clause 141 when directly or indirectly dependent from clause 59 wherein the inlet end of the first one of the valve apparatus of the vacuum pump system is connected to the waste container at the container outlet of the waste container. [0426] 143. The system of clause 142 when directly or indirectly dependent from clause 60 wherein the channel inlet clamp of the first one of the valve apparatus of the vacuum pump system releasably secures the inlet end of the first one of the valve apparatus of the vacuum pump system to the waste container at the container outlet of the waste container. [0427] 144. The system of clause 142 or 143 when directly or indirectly dependent from clause 61 wherein the waste container further comprises at least one waste container outlet interlocking portion at the container outlet, the at least one waste container outlet interlocking portion releasably interlocked with a corresponding one of the at least one inlet end interlocking portion of the inlet end of the first one of the valve apparatus to releasably secure the inlet end of the first one of the valve apparatus to the waste container at the container outlet of the waste container. [0428] 145. The system of clause 142, 143, or 144 when directly or indirectly dependent from clause 66 wherein the inlet end mating surface of the first one of the valve apparatus of the vacuum pump system engages with a corresponding container outlet mating surface of the waste container. [0429] 146. The system of clause 145 when directly or indirectly dependent from clause 67 wherein the channel inlet seal of the first one of the valve apparatus of the vacuum pump system seals an interface between the inlet end mating surface of the first one of the valve apparatus of the vacuum pump system and the container outlet mating surface of the waste container. [0430] 147. The system of clause 133 when directly or indirectly

dependent from clause 126 wherein:

[0431] the at least one wall of the waste container defines a container inlet for receiving the waste into the waste container; and

[0432] the flow channel of the second one of the valve apparatus of the vacuum pump system is in fluid communication with the chamber of the waste container through the container inlet of the waste container and through the channel outlet of the second one of the valve apparatus of vacuum pump system. [0433] 148. The system of clause 147 when directly or indirectly dependent from clause 72 wherein the outlet end of the second one of the valve apparatus of the vacuum pump system is connected to the waste container at the container inlet of the waste container. [0434] 149. The system of clause 148 when directly or indirectly dependent from clause 73 wherein the channel outlet clamp of the second one of the valve apparatus of the vacuum pump system releasably secures the outlet end of the second one of the valve apparatus of the vacuum pump system to the waste container at the container inlet of the waste container. [0435] 150. The system of clause 148 or 149 when directly or indirectly dependent from clause 74 wherein the waste container further comprises at least one waste container inlet interlocking portion at the container inlet, the at least one waste container inlet interlocking portion releasably interlocked with a corresponding one of the at least one outlet end interlocking portion of the outlet end of the second one of the valve apparatus to releasably secure the outlet end of the second one of the valve apparatus to the waste container at the container inlet of the waste container. [0436] 151. The system of clause 148, 149, or 150 when directly or indirectly dependent from clause 79 wherein the outlet end mating surface of the second one of the valve apparatus of the vacuum pump system engages with a corresponding container inlet mating surface of the waste container. [0437] 152. The system of clause 151 when directly or indirectly dependent from clause 80 wherein the channel outlet seal of the second one of the valve apparatus of the vacuum pump system seals an interface between the outlet end mating surface of the second one of the valve apparatus of the vacuum pump system and the container inlet mating surface of the waste container. [0438] 153. The system of any one of clauses 133 to 152 wherein the waste container comprises the waste container of any one of clauses 1 to 48. [0439] 154. A kit comprising:

[0440] the waste container of any one of clauses 1 to 48; and

[0441] at least one of the valve apparatus of any one of clauses 56 to 93. [0442] 155. The kit of clause 154 further comprising a pump. [0443] 156. The kit of clause 155 wherein the pump comprises a vacuum pump. [0444] 157. The kit of clause 155 or 156 wherein the at least one of the valve apparatus of any one of clauses 56 to 93 comprises at least two of the valve apparatus of any one of clauses 56 to 93.

[0445] Although specific embodiments have been described and illustrated, such embodiments should be considered illustrative only and not as limiting the invention as construed according to the accompanying claims.

Claims

1. A valve apparatus for controlling flow of waste into or out of a waste container, the valve apparatus comprising: a non-dismantlable valve body defining a flow channel for conveying the waste, the flow channel comprising a channel inlet for receiving the waste into the flow channel and a channel outlet for removing the waste from the flow channel; and at least one check valve secured within the flow channel, the at least one check valve configured to: allow the waste to flow through the flow channel in a forward flow direction from the channel inlet to the channel outlet; and prevent the waste from flowing through the flow channel in a reverse flow direction from the channel outlet to the channel inlet.

2. The valve apparatus of claim 1 wherein the valve body comprises an inlet end defining the channel inlet, the inlet end connectable to a waste source to cause the flow channel to be in fluid

communication with the waste source through the channel inlet.

3. The valve apparatus of claim 2 wherein the inlet end comprises at least one inlet end interlocking portion, the at least one inlet end interlocking portion releasably interlockable with a corresponding waste source interlocking portion of the waste source to releasably secure the inlet end to the waste source to connect the inlet end to the waste source to cause the flow channel to be in fluid communication with the waste source through the channel inlet.

4. The valve apparatus of claim 3 wherein the at least one inlet end interlocking portion is releasably interlockable with the waste source interlocking portion by rotation of the valve apparatus relative to the waste source.

5. The valve apparatus of claim 4 wherein the rotation of the valve apparatus relative to the waste source is around an inlet channel axis parallel to the flow channel at the channel inlet.

6. The valve apparatus of claim 3 wherein the at least one inlet end interlocking portion projects from the valve body at the inlet end.

7. The valve apparatus of claim 6 wherein the at least one inlet end interlocking portion projects out from the valve body away from the flow channel.

8. The valve apparatus of claim 2 wherein the inlet end comprises an inlet end mating surface around the channel inlet, the inlet end mating surface configured to engage with a corresponding waste source mating surface of the waste source when the inlet end is connected to the waste source to cause the flow channel to be in fluid communication with the waste source through the channel inlet.

9. The valve apparatus of claim 1 wherein the valve body comprises an outlet end defining the channel outlet, the outlet end connectable to a waste destination to cause the flow channel to be in fluid communication with the waste destination through the channel outlet.

10. The valve apparatus of claim 9 wherein the outlet end comprises at least one outlet end interlocking portion, the at least one outlet end interlocking portion releasably interlockable with a corresponding waste destination interlocking portion of the waste destination to releasably secure the outlet end to the waste destination to connect the outlet end to the waste destination to cause the flow channel to be in fluid communication with the waste destination through the channel outlet.

11. The valve apparatus of claim 10 wherein the at least one outlet end interlocking portion is releasably interlockable with the waste destination interlocking portion by rotation of the valve apparatus relative to the waste destination.

12. The valve apparatus of claim 11 wherein the rotation of the valve apparatus relative to the waste destination is around an outlet channel axis parallel to the flow channel at the channel outlet.

13. The valve apparatus of claim 10 wherein the at least one outlet end interlocking portion projects from the valve body at the outlet end.

14. The valve apparatus of claim 13 wherein the at least one outlet end interlocking portion projects out from the valve body away from the flow channel.

15. The valve apparatus of claim 9 wherein the outlet end comprises an outlet end mating surface around the channel outlet, the outlet end mating surface configured to engage with a corresponding waste destination mating surface of the waste destination when the outlet end is connected to the waste destination to cause the flow channel to be in fluid communication with the waste destination through the channel outlet.

16. The valve apparatus of claim 1 wherein the valve body comprises a plurality of valve body portions between the channel inlet and the channel outlet, the plurality of valve body portions collectively defining the flow channel, each valve body portion of the plurality of valve body portions fixed to at least one other valve body portion of the plurality of valve body portions.

17. The valve apparatus of claim 16 wherein at least two of the valve body portions of the plurality of valve body portions are fixed together by plastic welding.

18. The valve apparatus of claim 1 wherein the at least one check valve comprises at least one duckbill valve.

- 19.** A method of installing the valve apparatus of claim 4 to the waste source, the method comprising: rotating the valve apparatus relative to the waste source to interlock the at least one inlet end interlocking portion with the waste source interlocking portion.
- 20.** A method of installing the valve apparatus of claim 11 to the waste destination, the method comprising: rotating the valve apparatus relative to the waste destination to interlock the at least one outlet end interlocking portion with the waste destination interlocking portion.
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